

Analysis of the Kinetics and Bistability of Ubiquinol:Cytochrome *c* Oxidoreductase

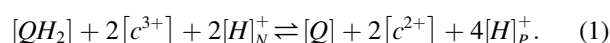
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ABSTRACT Ubiquinol:cytochrome *c* oxidoreductase, *bc*₁ complex, is the enzyme in the respiratory chain of mitochondria responsible for the transfer reducing potential from ubiquinol to cytochrome *c* coupled to the movement of charge against the electrostatic potential across the mitochondrial inner membrane. The complex is also implicated in the generation of reactive oxygen species under certain conditions and is thus a contributor to cellular oxidative stress. Here, a biophysically detailed, thermodynamically consistent model of the *bc*₁ complex for mammalian mitochondria is developed. The model incorporates the major redox centers near the Q_o- and Q_i-site of the enzyme, includes the pH-dependent redox reactions, accounts for the effect of the proton-motive force of the reaction rate, and simulates superoxide production at the Q_o-site. The model consists of six distinct states characterized by the mobile electron distribution in the enzyme. Within each state, substates that correspond to various electron localizations exist in a rapid equilibrium distribution. The steady-state equation for the six-state system is parameterized using five independent data sets and validated in comparison to additional experimental data. Model analysis suggests that the pH-dependence on turnover is primarily due to the pK_a values of cytochrome *b*_H and Rieske iron sulfur protein. A previously proposed kinetic scheme at the Q_i-site where ubiquinone binds to only the reduced enzyme and ubiquinol binds to only the oxidized enzyme is shown to be thermodynamically infeasible. Moreover, the model is able to reproduce the bistability phenomenon where at a given overall flux through the enzyme, different rates of superoxide production are attained when the enzyme is differentially reduced.

INTRODUCTION

Ubiquinol:cytochrome *c* oxidoreductase, *bc*₁ complex, is an inner membrane protein complex that catalyzes the reduction of two cytochrome (cyt) *c* molecules coupled to the overall oxidation of one ubiquinol molecule and effectively pumps two protons across the mitochondrial inner membrane (1). The enzyme utilizes a ubiquinone (Q)-cycle mechanism where the two-electron oxidation of ubiquinol is bifurcated so that one electron passes through the high-potential chain and the other down the low-potential chain. The high-potential chain consists of the Rieske iron sulfur protein, tightly bound cyt *c*₁ and the mobile electron carrier cyt *c*. The low-potential chain consists of the membrane spanning cyt *b* that includes cyt *b*_L and cyt *b*_H. One complete catalytic cycle requires the oxidation of two ubiquinol molecules and the reduction of one ubiquinone molecule back to ubiquinol. Therefore, it possesses both a ubiquinone oxidase and reductase. The oxidase is at the Q_o-site, which resides at the intermembrane space side (P-side) of the inner membrane and facilitates the sequential electron bifurcation reaction. The first electron is used to reduce cyt *c*, and the second electron is used to facilitate proton pumping via the reductase reaction. The reductase is at the Q_i-site at the opposite end of the enzyme (N-side). This is where ubiquinol is regenerated upon the second turnover of the Q_o-site. The biochemical equation for the net reaction is shown as



The *bc*₁ complex is also known to produce superoxide (2–4). Superoxide is an important signaling molecule at physiological concentrations (5–7), but it can lead to significant oxidative-induced damage at elevated patho-physiological levels (for review, see Zorov et al. (8)). Under normal conditions, in vitro superoxide production is low (0.12–0.8% of total oxygen consumption (9–11)). However, when the proton-motive backpressure is large, the Q-pool becomes highly reduced or in the presences of a Q_i-site inhibitor (e.g., antimycin A), the superoxide production rate can significantly increase. The putative mechanism of superoxide production is via a one-electron reduction of molecular oxygen by an unstable semiquinone (SQ) at the Q_o-site (12), but the precise nature of this side reaction is obscure (13).

Although several models of the *bc*₁ complex have been developed (14–24), they are either too simple to provide insight into the catalytic cycle or too complex to be confidently identified. In addition to investigating the catalytic mechanism, we seek a suitable model to integrate into the contemporary generation of mitochondrial bioenergetics models and simulate superoxide production (15,25,26). Most simple models are of the ping-pong type and do not include thermodynamic constraints and superoxide-producing mechanisms (16–20). Other models include those that are thermodynamically balanced, mass-action based (14,15), and a Q-pool-based mechanism assuming substrate saturation at the Q_i-site (21). Complex models use

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high-dimensional systems of ordinary differential equations (ODEs) to describe the enzyme kinetics (22–24). One of these is a 13-state model that simulates the oxidative states for several of the redox centers and accounts for a number of superoxide production pathways (22). The other two high-dimensional models include more oxidative states in 56-state (23) and 256-state (24) systems of ODEs. The 256-state ODE model also includes the superoxide production pathways and simulates the bistability phenomenon, where different rates of superoxide production occur at a given flux through the net reaction (Eq. 1). This phenomenon may be important in the pathogenesis of ischemia-reperfusion injury (24,27).

The model developed here maintains the features presented above in a tenable, well-constrained representation of the bc_1 complex. It is a parsimonious representation of the catalytic cycle that explains a wide variety of independent data sets and incorporates all the major redox centers near the Q_o - and Q_i -site of the complex, maintains all the pH-dependent redox reactions, accounts for the effect of the proton-motive force, includes the putative superoxide producing pathways, and reproduces the bistability phenomenon.

METHODS

Model structure

The scheme of the catalytic cycle and state model diagram of the bc_1 complex model are shown in Fig. 1. The schematic for the overall reaction is shown in Fig. 1 A. The model includes the redox biochemistry that occurs at the Q_o -site and Q_i -site of the complex and couples cyt c reduction with the first electron transfer from ubiquinol. The Q_o -site includes the ubiquinone oxidase binding site, iron sulfur protein (ISP), and cyt b_L ; the Q_i -site includes cyt b_H and the ubiquinone reductase binding site. It is assumed that the reactions at each site are independent from each other. This first

electron transfer at the Q_o -site is the one of the primary, rate-limiting steps in the catalytic cycle (28). During steady-state turnover, we assume that up to two mobile electrons can exist at both the Q_o -site and Q_i -site, which depends on the reducing environment and the proton-motive force. (The proton motive-force is the chemical potential associated with moving a proton from the N-side to the P-side of the membrane, and is sometimes termed a backpressure.) All these combinations result in six distinct states that characterize the electron distribution in the enzyme as shown in Fig. 1 B. Within each state, substates exist in rapid equilibrium with each other and represent different electron localizations at each site. State transitions are only allowed to occur from certain substates. Some possible substate combinations are forbidden. For example, a ubiquinone cannot be bound at the Q_o -site for states E5 and E6. It is assumed that an unstable SQ is occupies the ubiquinone binding site. For details concerning the state transition-rates, see Primary state transitions below.

Midpoint potentials

The fractional substate occupancies and the state transitions are governed by the thermodynamic driving force of the redox biochemistry defined by the midpoint potentials. The midpoint potentials (with respect to pH 0) are given in Table 1. Many of the redox centers exhibit redox-linked protonations and thus have pH-dependent midpoint potentials. The pH corrected values are computed using:

$$E_m\left(\frac{Q}{SQ_o}\right) = E_m^0\left(\frac{Q}{QH_2}\right) + \frac{RT}{2F} \ln(K_{\text{stabo}}), \quad (2)$$

$$E_m\left(\frac{SQ_o}{QH_2}\right) = 2E_m^0\left(\frac{Q}{QH_2}\right) - E_m^0\left(\frac{Q}{SQ_o}\right) + \frac{RT}{F} \ln([H^+]_P^2), \quad (3)$$

$$E_m\left(\frac{b_L^{3+}}{b_L^{2+}}\right) = E_m^0\left(\frac{b_L^{3+}}{b_L^{2+}}\right) - \frac{RT}{F} \ln\left(\frac{([H^+]_P + 10^{-pK_{bL,ox}})}{([H^+]_P + 10^{-pK_{bL,red}})}\right), \quad (4)$$

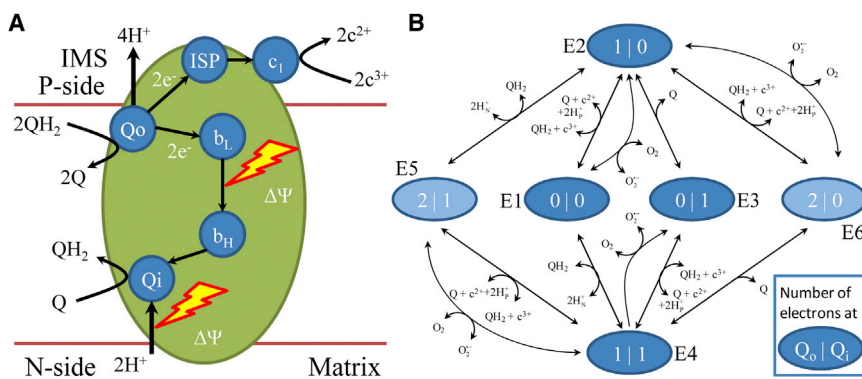


FIGURE 1 Model reaction schematic and state model representation for the bc_1 complex. (A) The overall reaction is shown where two cyt c values are reduced for a net oxidation of one ubiquinol molecule and two protons are pumped across the inner-mitochondrial membrane. The P-side is the positive or IMS side of the membrane and the N-side is the negative or matrix side of the membrane. Electron transfer from cyt b_L to cyt b_H and proton uptake at the N-side are assumed to be electrogenic. The Q_o -site includes the ubiquinone oxidase binding site, ISP, and cyt b_L ; the Q_i -site includes cyt b_H and the ubiquinone reductase binding site. (B) The six-state model used to simulate the kinetics of the bc_1 complex is shown. (Ovals)

Enzyme states and their mobile electron distributions. (Directional arrows) State transitions constituting the catalytic mechanism. The partial reactions associated with the directional arrows are the typical reactions associated with the state transitions. For example, ability of ubiquinone to bind to the Q_i -site to form a stable SQ via a sequential reduction by cyt b_L with cyt b_H as the intermediary is reflected in the transition from E2 to E3 (or E6 to E4). However, ubiquinone does not have to bind at the Q_i -site for this transition to occur. The only requirement for this transition is that cyt b_H is oxidized. But ubiquinone must be bound before the transition from E4 to E1 (or E5 to E2) so that two mobile electrons can reside at the Q_i -site (one on cyt b_H and the other on the stable SQ) and lead to the formation of ubiquinol. As such, a catalytic cycle results in net electron transfer from ubiquinol to oxidized cyt c , which may be represented by the primary state transition sequence E1-E2-E3-E4-E1. The numbers represent the number of electrons at the Q_o -site on the left of the vertical line and the Q_i -site on the right. For states E1, E2, and E6, the Q_i -site can have either 0 or two electrons indicated by 0/2. (Dark ovals) The four primary states. (Light ovals) The two extra states that are mainly involved in superoxide production. See the Methods for further details.

TABLE 1 Model parameters

Parameter	Definition	Value	Reference
R	Ideal gas constant	8.314 J/mol/K	—
F	Faraday's constant	96.4 J/mol/mV	—
$E_m(c^{3+}/c^{2+})$	Cytochrome c midpoint potential	230 mV	(61)
$E_m^0(Q/QH_2)$	Ubiquinone midpoint potential	477 mV	(30) ^a
$E_m^0(b_L^{3+}/b_L^{2+})$	Cytochrome b_L midpoint potential	39 mV	(31)
$E_m^0(b_H^{3+}/b_H^{2+})$	Cytochrome b_H midpoint potential	160 mV	(31)
$E_m(O_2/O_2^{\cdot-})$	Superoxide midpoint potential	−160 mV	(62) ^b
K_{stabo}	Q _o -site semiquinone stability constant	10^{-33} unitless	(63) ^{b,c}
K_{stabi}	Q _i -site semiquinone stability constant	$10^{-16.11}$ unitless	(30) ^b
pK_{ISPox1}	Oxidized Rieske ISP acidic pK	7.6	(39)
pK_{ISPox2}	Oxidized Rieske ISP alkaline pK	9.2	(39)
pK_{bLox}	Oxidized cytochrome b_L pK	5.9	(31)
pK_{bLred}	Reduced cytochrome b_L pK	7.9	(31)
pK_{bHox}	Oxidized cytochrome b_H pK	5.7	(31)
pK_{bHred}	Reduced cytochrome b_H pK	7.7	(31)
β	Fractional charge transfer coefficient	0.5 unitless	(31,34–36)
k_{SO}	First order superoxide production-rate constant	8000 s ^{−1}	(63)
k_{340}	State E3 to E4 intrinsic transition-rate constant	6.2×10^3 s ^{−1}	— ^d
k_{260}	State E2 to E6 intrinsic transition-rate constant	6.2×10^3 s ^{−1}	— ^d
k_{640}	State E6 to E4 intrinsic transition-rate constant	3.3×10^7 s ^{−1}	— ^e
k_{450}	State E4 to E5 intrinsic transition-rate constant	3 s ^{−1}	— ^f
k_{520}	State E5 to E2 intrinsic transition-rate constant	3.2×10^4 s ^{−1}	— ^g

^aBased on extrapolating from 60 mV at pH 7 and 25°C to pH 0.

^bExtrapolated to pH 0 from given reference. At pH 7, $K_{\text{stabo}} = 10^{-9}$ and $K_{\text{stabi}} = 10^{-2.11}$.

^cBased on approximate K_{eq} for the superoxide reaction.

^dIdentical state transition as E1 to E2.

^eIdentical state transition as E2 to E3.

^fThis rate must be low to simulate physiological superoxide production rates.

^gIdentical state transition as E4 to E1.

$$E_m\left(\frac{b_H^{3+}}{b_H^{2+}}\right) = E_m^0\left(\frac{b_H^{3+}}{b_H^{2+}}\right) - \frac{RT}{F} \ln\left(\frac{([H^+]_N + 10^{-pK_{bHox}})}{([H^+]_N + 10^{-pK_{bHred}})}\right), \quad (5)$$

$$E_m\left(\frac{Q}{SQ_i}\right) = E_m^0\left(\frac{Q}{QH_2}\right) + \frac{RT}{2F} \ln(K_{\text{stabi}}), \quad (6)$$

$$E_m\left(\frac{SQ_i}{QH_2}\right) = 2E_m^0\left(\frac{Q}{QH_2}\right) - E_m^0\left(\frac{Q}{SQ_i}\right) + \frac{RT}{F} \ln([H^+]_N^2). \quad (7)$$

For the ubiquinone midpoint potentials, Equations 2, 3, 6, and 7 are good approximations when pH is between 6 and 10 assuming a pK_a of 4.9 for the SQ and pK_a values of 11.3 and 13.2 for ubiquinol (29). These approximations are further supported by experimental measurements of the midpoint potentials that demonstrate the pH-independence of the Q/SQ couple and a −120 mV/pH dependence of the SQ/ubiquinol (QH₂) couple (30). The stability constants used in Eqs. 2 and 6 are with respect to pH 0. The pK values for cyt b_L and cyt b_H were obtained from Rich et al. (31).

Reactant binding

It is assumed that substrates, products, and protons bind to the *bc*₁ complex much more rapidly than the state transition rates. As such, binding polynomial expressions are used to compute the fraction that a given state is bound with a particular reactant. A binding polynomial is a partition function that is used to determine the fraction a binding site is bound with a given ligand (32). These binding polynomials for the Q_o-site, cyt c binding site, Q_i-site, and the protonated state of the Rieske ISP are shown in Eqs. 8–11. As an example, the binding polynomial for the Q_o-site partitions this binding site into free, ubiquinol-bound or ubiquinone-bound fractions. As such, the term $[QH_2]/K_{QH_2}^o/P_{Qo}$ defines how much ubiquinol is bound to the enzyme at the Q_o-site in a given allowable state where $[QH_2]$ is the concentration of free ubiquinol, $K_{QH_2}^o$ is the dissociation constant for ubiquinol at the Q_o-site, and P_{Qo} is the binding polynomial for the Q_o-site,

$$P_{Qo} = 1 + \frac{[QH_2]}{K_{QH_2}^o} + \frac{[Q]}{K_Q^o}, \quad (8)$$

$$P_c = 1 + \frac{[c^{2+}]}{K_{c2}} + \frac{[c^{3+}]}{K_{c3}}, \quad (9)$$

$$P_{Qi} = 1 + \frac{[QH_2]}{K_{QH_2}^i} + \frac{[Q]}{K_Q^i}, \quad (10)$$

$$P_{\text{ISP}} = 1 + \frac{[H^+]_P}{10^{-pK_{\text{ISPox2}}}} + \frac{[H^+]_P^2/10^{-pK_{\text{ISPox1}}}}{10^{-pK_{\text{ISPox1}}}}. \quad (11)$$

Substates

When there is one electron at the Q_o-site, it is able to rapidly localize to an unstable SQ or cyt b_L . However, it resides on cyt b_L >99% of the time due to the very low reduction potential of the unstable SQ. Under the right circumstances when cyt b_L is reduced, there is a small but significant level of SQ at the Q_o-site. Equations 12–15 are the substate equations for the Q_o-site. The terms $1/f_{Qo}$ and r_{Qo}/f_{Qo} are the fractions of a given state where the electron at the Q_o-site resides on an unstable SQ or cyt b_L , respectively. The two other substates for the Q_o-site are when there are zero or two (on an unstable SQ and reduced cyt b_L) electrons present,

$$\Delta G_{Qo} = -F\left(E_m\left(\frac{b_L^{3+}}{b_L^{2+}}\right) - E_m\left(\frac{Q}{SQ_o}\right)\right), \quad (12)$$

$$K_{Qo} = e^{-\Delta G_{Qo}/RT}, \quad (13)$$

$$r_{Qo} = \frac{K_{Qo}P_{Qo}K_Q^o}{[Q]}, \quad (14)$$

$$f_{Qo} = 1 + r_{Qo}. \quad (15)$$

When there is one electron at the Q_i -site, it is able to rapidly localize to cyt b_H or a stable SQ. The substate equations for this condition are shown in Eqs. 16–19. The terms $1/f_{Qi,1}$ and $r_{Qi,1}/f_{Qi,1}$ are the fractions of a given state where the electron at the Q_i -site resides on cyt b_H or a stable SQ, respectively,

$$\Delta G_{Qi,1} = -F \left(E_m \left(\frac{Q}{SQ_i} \right) - E_m \left(\frac{b_H^{3+}}{b_H^{2+}} \right) \right), \quad (16)$$

$$K_{Qi,1} = e^{-\Delta G_{Qi,1}/RT}, \quad (17)$$

$$r_{Qi,1} = \frac{K_{Qi,1}[Q]/P_{Qi}}{K_Q^i}, \quad (18)$$

$$f_{Qi,1} = 1 + r_{Qi,1}. \quad (19)$$

When there are two electrons at the Q_i -site, they are on cyt b_H and the stable SQ; however, reduction of the stable SQ by cyt b_H regenerates ubiquinol and vacates the two electrons from the Q_i -site. It is assumed that proton uptake from the N-side, mitochondrial matrix, is coupled to the formation of the ubiquinol anion when the stable SQ is reduced. This process is dependent on the proton-motive force. Equations 20–23 are the substate equations for the Q_i -site when there are either two or zero electrons present. The terms $1/f_{Qi,2}$ and $r_{Qi,2}/f_{Qi,2}$ are the fractions of a given state where the electrons at the Q_i -site reside on cyt b_H and a stable SQ or there are no electrons present, respectively. The parameter, β , is the effective fractional distance that charge is translocated through the membrane potential,

$$\Delta G_{Qi,2} = -F \left(E_m \left(\frac{SQ_i}{QH_2} \right) - E_m \left(\frac{b_H^{3+}}{b_H^{2+}} \right) \right) + 2(1 - \beta)F\Delta\Psi, \quad (20)$$

$$K_{Qi,2} = e^{-\Delta G_{Qi,2}/RT}, \quad (21)$$

$$r_{Qi,2} = \frac{K_{Qi,2}K_{QH_2}^i P_{Qi}}{[QH_2]}, \quad (22)$$

$$f_{Qi,2} = 1 + r_{Qi,2}. \quad (23)$$

Primary state transitions

State transitions are governed by two primary Gibbs's free energies of reaction. At the Q_o -site, the thermodynamic driving force for the first electron oxidation of ubiquinol is set by the midpoint potential difference between cyt c and the unstable SQ. This governs the state transitions from state E1 to E2, E3 to E4, E4 to E5, and E2 to E6. For simplicity, the mobile electron distribution in the high-potential chain (ISP and cyt c_1) is not considered. At the Q_i -site, the driving force for the electron transfer from cyt b_L to cyt b_H governs the state transition between states E2 to E3, E4 to E1, E5 to E2, and E6 to E4. These two driving forces are computed using

$$\Delta G_o = -F \left(E_m \left(\frac{c^{3+}}{c^{2+}} \right) - E_m \left(\frac{SQ_o}{QH_2} \right) \right), \quad (24)$$

$$\Delta G_i = -F \left(E_m \left(\frac{b_H^{3+}}{b_H^{2+}} \right) - E_m \left(\frac{b_L^{3+}}{b_L^{2+}} \right) \right). \quad (25)$$

A third Gibbs's free energy of reaction is used for the superoxide producing pathways. Superoxide production causes states to transition from E2 to E1, E4 to E3, E5 to E4, or E6 to E2. This driving force is computed using

$$\Delta G_{SO} = -F \left(E_m \left(\frac{O_2}{O_2^-} \right) - E_m \left(\frac{Q}{SQ_o} \right) \right). \quad (26)$$

State E1 includes substates characterizing 0 electrons at the Q_o -site and either two electrons at the Q_i -site (one on cyt b_H and the other on a stable SQ) or the fully oxidized enzyme (both the Q_o -site and Q_i -site contain 0 electrons). For simplicity, we will step through and describe the catalytic cycle, assuming it is fully oxidized, and loop around the four primary states for the forward reaction (*dark ovals* in Fig. 1 B). The transition from state E1 to E2 is rate-limiting due to the existence of a high activation-energy barrier (28,33) and not addressed here. For simplicity, the first electron transfer from ubiquinol oxidation is coupled to the reduction of cyt c through the high-potential chain. The second electron is then transferred from an unstable SQ to cyt b_L . The transition from state E2 to E3 moves a negative charge partially across the inner membrane from cyt b_L to cyt b_H and ultimately to ubiquinone at the Q_i -site to form a stable SQ. This step is dependent on the electrostatic potential across the membrane, $\Delta\Psi$, and is only allowed to occur when cyt b_H is oxidized. (Potential difference $\Delta\Psi$ is defined as the P-side potential minus the N-side potential.) The Q_o -site turns over again to transition from state E3 to E4 in the same manner as the transition from state E1 to E2. State E4 has one electron at the Q_o -site and one electron at the Q_i -site. To transition from state E4 to E1, the electron at the Q_o -site must be on cyt b_L , and cyt b_H must be oxidized (the electron at the Q_i -site is on the stable SQ). After state E4 transitions to E1, the stable SQ at the Q_i -site is reduced to the ubiquinol anion, two matrix protons are taken up against an electrochemical potential, ubiquinol is released (regenerated), and the enzyme is ready to catalyze another cycle. This scenario represents one possible sequence of events during a catalytic cycle. For example, depending on the reducing environment and proton-motive force backpressure, ubiquinol may not be regenerated until before the transition from state E2 to E3. The state transition rates for these primary four states are given in

$$k_{12} = k_{120} \frac{[QH_2]}{K_{QH_2}^o P_{Qo}} \frac{[c^{3+}][H^+]_P (1 + [H^+]_P / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} + k_{SO} e^{\Delta G_{SO}/RT} \frac{[O_2^-][Q]}{K_Q^o P_{Qo}}, \quad (27)$$

$$k_{21} = k_{120} e^{\Delta G_o/RT} \frac{[c^{2+}]}{K_{c2} P_c f_{Qo}} \frac{1}{10^{-pK_{ISPox2}} P_{ISP}} \frac{[H^+]_P (1 + [H^+]_P / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} + k_{SO} \frac{[O_2]}{f_{Qo}}, \quad (28)$$

$$k_{23} = k_{230} \frac{r_{Qo} r_{Qi,2}}{f_{Qo} f_{Qi,2}} e^{-\beta F \Delta\Psi / 2RT}, \quad (29)$$

$$k_{32} = k_{230} e^{\Delta G_i/RT} \frac{e^{\beta F \Delta\Psi / 2RT}}{f_{Qi,1}}, \quad (30)$$

$$k_{34} = k_{340} \frac{[QH_2]}{K_{QH_2}^o P_{Qo}} \frac{[c^{3+}][H^+]_P (1 + [H^+]_P / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} + k_{SO} e^{\Delta G_{SO}/RT} \frac{[O_2^-][Q]}{K_Q^o P_{Qo}}, \quad (31)$$

$$k_{43} = k_{340} e^{\Delta G_o/RT} \frac{[c^{2+}]}{K_{c2} P_c f_{Qo}} \frac{1}{10^{-pK_{ISPox2}} P_{ISP}} \frac{[H^+]_P (1 + [H^+]_P / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} + k_{SO} \frac{[O_2]}{f_{Qo}}, \quad (32)$$

$$k_{41} = k_{410} \frac{r_{Qo} r_{Qi,1}}{f_{Qo} f_{Qi,1}} e^{-\beta F \Delta \Psi / 2RT}, \quad (33)$$

$$k_{14} = k_{410} e^{\Delta G_i/RT} \frac{e^{\beta F \Delta \Psi / 2RT}}{f_{Qi,2}}. \quad (34)$$

All intrinsic rate constants for the reverse reaction are defined according to mass action: $k_r = k_f e^{\Delta G/RT}$, where k_r and k_f are the reverse and forward rate constants, respectively, and $\Delta G/RT$ is the unitless Gibbs free energy for the reaction. A detailed example describing how the transition rates are derived is located in the [Supporting Material](#).

Auxiliary state transitions

In a highly reduced environment and/or in the presence of a large proton-motive force, the enzyme will transition into states E5 and E6 (*light ovals* in [Fig. 1 B](#)); however, under normal operating conditions, the fractional occupancy of these states is small relative to the primary states. In these auxiliary states, superoxide may form when the unstable SQ at the Q_o-site reacts with oxygen. The state-transition rates for these states are given in

$$k_{26} = k_{260} \frac{[QH_2]}{K_{QH2}^o P_{Qo}} \frac{[c^{3+}]}{K_{c3} P_c f_{Qo}} \frac{r_{Qo} [H^+]_P (1 + [H^+]_P / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} + k_{SO} e^{\Delta G_{SO}/RT} \frac{[O_2^-][Q]}{K_Q^o P_{Qo} f_{Qo}}, \quad (35)$$

$$k_{62} = k_{260} e^{\Delta G_o/RT} \frac{[c^{2+}]}{K_{c2} P_c} \frac{[H^+]_P (1 + [H^+]_P / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} + k_{SO} [O_2], \quad (36)$$

$$k_{64} = k_{640} \frac{r_{Qi,2}}{f_{Qi,2}} e^{-\beta F \Delta \Psi / 2RT}, \quad (37)$$

$$k_{46} = k_{640} e^{\Delta G_i/RT} \frac{e^{\beta F \Delta \Psi / 2RT}}{f_{Qo} f_{Qi,1}}, \quad (38)$$

$$k_{45} = k_{450} \frac{[QH_2]}{K_{QH2}^o P_{Qo}} \frac{[c^{3+}]}{K_{c3} P_c f_{Qo}} \frac{r_{Qo} [H^+]_P (1 + [H^+]_P / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} + k_{SO} e^{\Delta G_{SO}/RT} \frac{[O_2^-][Q]}{K_Q^o P_{Qo} f_{Qo}}, \quad (39)$$

$$k_{54} = k_{450} e^{\Delta G_o/RT} \frac{[c^{2+}]}{K_{c2} P_c} \frac{[H^+]_P (1 + [H^+]_P / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} + k_{SO} [O_2], \quad (40)$$

$$k_{52} = k_{520} \frac{r_{Qi,1}}{f_{Qi,1}} e^{-\beta F \Delta \Psi / 2RT}, \quad (41)$$

$$k_{25} = k_{520} e^{\Delta G_i/RT} \frac{e^{\beta F \Delta \Psi / 2RT}}{f_{Qo} f_{Qi,2}}. \quad (42)$$

Flux expressions

The net turnover flux through the enzyme is computed by summing the net fluxes through states E1 and E2 and states E5 and E2 as shown in Eq. 43. In the steady state, mass conservation requires that the sum of these two net fluxes must equal the sum of the net fluxes through states E2 and E6 and states E2 and E3, and thus the net reaction flux,

$$J_{bc1} = E_{tot} (k_{12} E1 + k_{52} E5 - E2 (k_{21} + k_{25})). \quad (43)$$

The superoxide production rate is computed by summing the net superoxide production that occurs when the unstable SQ reacts with oxygen as shown in Eq. 44. For simplicity, the reverse rate, superoxide oxidation, is not shown. It is negligible compared to superoxide production,

$$J_{ROS} = E_{tot} k_{SO} [O_2] \left(E5 + E6 + \frac{(E2 + E4)}{f_{Qo}} \right). \quad (44)$$

The analytic expressions for the states at steady state are obtained by solving the system of equations presented in Eq. 45. The first six rows correspond to the state equations governing the system at steady state. The last row is used to set the steady-state solutions to fractional occupancies (i.e., $\sum_i E_i = 1$). The analytical solution for each state contains ~1000 terms, so they are not explicitly presented here:

$$\begin{pmatrix} -(k_{12} + k_{14}) & k_{21} & 0 & k_{41} & 0 & 0 \\ k_{12} & -(k_{21} + k_{23} + k_{25} + k_{26}) & k_{32} & 0 & k_{52} & k_{62} \\ 0 & k_{23} & -(k_{32} + k_{34}) & k_{43} & 0 & 0 \\ k_{14} & 0 & k_{34} & -(k_{41} + k_{43} + k_{45} + k_{46}) & k_{54} & k_{64} \\ 0 & k_{25} & 0 & k_{45} & -(k_{52} + k_{54}) & 0 \\ 0 & k_{26} & 0 & k_{46} & 0 & -(k_{62} + k_{64}) \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{bmatrix} E1 \\ E2 \\ E3 \\ E4 \\ E5 \\ E6 \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (45)$$

The model was developed, parameterized, and simulated on a Dell Precision T3500 workstation with a 3.2 GHz Intel Xeon quadcore processor and 16 GB RAM using MATLAB ver. 2011b (The MathWorks, Natick, MA). The steady-state equation for the six-state model was solved analytically using MATLAB's symbolic toolbox. A custom, parallelized simulated annealing algorithm was used to globally search the parameter space before identifying a local minimum with a gradient-based local optimizer. For details concerning the experimental data and fitting and sensitivity analysis, see the [Supporting Material](#).

RESULTS AND DISCUSSION

Table 1 shows the fixed parameter values obtained from the literature and used to simulate the model. They consist of midpoint potentials, stability constants, pKa values, and other parameters. One of these parameters, β , is an important parameter defining how the internal electron transfer steps contribute to $\Delta\Psi$ -generation. There is strong experimental evidence suggesting that the reoxidation of cyt b_H results in half a charge transfer across the membrane (31,34–36). Previous models underestimated (24), overestimated (22), or did not address (14–21,23) this parameter. **Table 1** also includes the intrinsic state transition rates for the second turnover at the Q_o -site and the auxiliary states. It is assumed that the second ubiquinol is oxidized as the same rate as the first. The intrinsic rate constants for the

auxiliary states were set equal to their primary state counterparts except for the transition from state E4 to E5. This transition rate is required to be low to prevent unphysiological superoxide production rates.

Table 2 shows the fitted parameter values, the normalized sensitivity coefficients, and the sensitivity rankings. These parameters were fitted to five independent data sets (16,18,19,33,37) as shown in [Figs. 2–6](#). Turnover is computed as mol cyt c reduced per mol cyt c_1 per s as defined in the data sets. In all the data sets, horse heart cyt c was used as the oxidant. The top five sensitive parameters were two Q_o -site ubiquinol binding constants, one Q_i -site ubiquinone binding constant, and the intrinsic state transition-rate constants for states E1 to E2 and E4 to E1. As expected, each of these parameters is related to the rate-limiting step in the catalytic cycle. Although only two of the rate constants appear in the top-10-ranked sensitive parameters, they all are in excellent agreement with previous estimates of these values based on pre-steady-state kinetic measurements (38). Moreover, the majority of the ubiquinone binding constants are also in agreement with previous estimates. These are discussed in more detail below. Also, all the fitted Q_{ratio} values fall in expected ranges (0.1 to <5%) when the degree of nonenzymatic ubiquinol oxidation is assumed to be 1% of the average enzymatic rate.

TABLE 2 Fitted model parameters

Parameter	Definition	Value	Sensitivity	Rank
k_{120}	State E1 to E2 intrinsic transition-rate constant	$6.2 \times 10^3 \text{ s}^{-1}$	0.60	2
k_{230}	State E2 to E3 intrinsic transition-rate constant	$3.3 \times 10^7 \text{ s}^{-1}$	4.6×10^{-5}	23
k_{410}	State E4 to E1 intrinsic transition-rate constant	$3.2 \times 10^4 \text{ s}^{-1}$	0.40	5
K_{c3}	Cytochrome c^{3+} binding constant	1.1 μM	0.28	14
K_{c2}	Cytochrome c^{2+} binding constant	1.2 μM	0.21	18
$K_{c3,Mg}$	Mg^{2+} altered cytochrome c^{3+} binding constant	23 μM^a	0.22	16
$K_{c2,Mg}$	Mg^{2+} altered cytochrome c^{2+} binding constant	8.8 μM^a	0.17	20
K_{DQH2}^o	DQH ₂ Q_o -site binding constant	6.9 μM	0.62	1
K_{DQ}^o	DQ Q_o -site binding constant	0.01 μM	0.32	11
K_{DQ}^i	DQH ₂ Q_i -site binding constant	0.03 μM	0.17	21
K_{DQH2}^i	DQ Q_i -site binding constant	10.4 mM	0	— ^b
K_{NBH}^o	NBH Q_o -site binding constant	49 μM	0.42	4
K_{NB}^o	NB Q_o -site binding constant	0.32 μM	0.29	13
K_{NB}^i	NBH Q_i -site binding constant	0.01 μM	5.3×10^{-3}	22
K_{NBH}^i	NB Q_i -site binding constant	200 μM	0	— ^b
K_{Q2H2}^o	Q ₂ H ₂ Q_o -site binding constant	15 μM	0.19	19
K_{Q2}^o	Q ₂ Q_o -site binding constant	3.2 μM	0.30	12
K_{Q2}^i	Q ₂ H ₂ Q_i -site binding constant	0.25 μM	0.22	17
K_{Q2H2}^i	Q ₂ Q_i -site binding constant	4.7 μM	0	— ^b
K_{QH2}^o	Q ₂ H ₂ /Q ₁₀ H ₂ Q_o -site binding constant	0.08 μM	0.34	9
K_{Q}^o	Q ₂ /Q ₁₀ Q_o -site binding constant	0.05 μM	0.35	8
K_{Q}^i	Q ₂ H ₂ /Q ₁₀ H ₂ Q_i -site binding constant	13 μM	0.56	3
K_{QH2}^i	Q ₂ /Q ₁₀ Q_i -site binding constant	25 μM	0	— ^b
Q_{ratio1}^c	Percent initial ubiquinol oxidized	0.33%	0.34	10
Q_{ratio2}^c	Percent initial ubiquinol oxidized	0.99%	0.36	7
Q_{ratio3}^c	Percent initial ubiquinol oxidized	3.3%	0.27	15
Q_{ratio4}^c	Percent initial ubiquinol oxidized	0.63%	0.37	6

Q₁₀H₂, ubiquinol-10.

^aFor the fourth data set (18), the cytochrome c binding affinities were purposefully lowered by including 20 mM MgCl₂ in the reaction buffer.

^bDefined via microscopic reversibility where $K_{QH2}^i = K_{QH2}^o K_{Q}^i K_{c3}^2 / K_{Q2}^o / K_{c2}^2$.

^cNumbers correspond to the following data sets: 1 (19); 2 (33); 3 (16); and 4 (18).

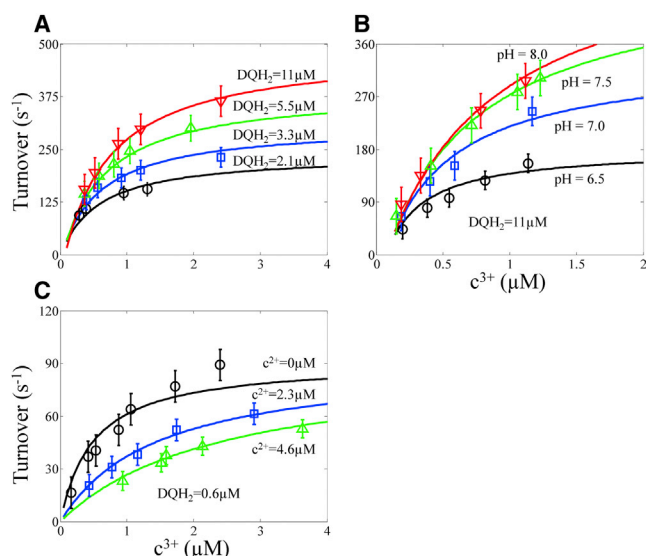


FIGURE 2 Turnover rate of isolated *bc*₁ complex oxidizing decylhydroquinone (DQH₂) (19). (A) The simulated rate of cyt *c* reduction versus c^{3+} concentration at various fixed DQH₂ concentrations. (B) The simulated rate of cyt *c* reduction versus c^{3+} concentration at various fixed pH at a fixed DQH₂ concentration. (C) The simulated rate of cyt *c* reduction versus c^{3+} concentration at various fixed c^{2+} concentrations at a fixed DQH₂ concentration. (Symbols) Experimental data. (Lines) Model simulations.

In the first data set, decylhydroquinone was used to explore the kinetics of isolated bovine *bc*₁ complex with varying pH conditions and ferricytochrome *c* (c^{3+}) concentrations including end-product inhibition by ferrocytochrome *c* (c^{2+}) (19). Fig. 2 demonstrates that the model is able to reproduce the data very well. The second data set shows the effect of pH on turnover of reconstituted bovine *bc*₁ complex oxidizing nonyl-ubihydroquinone and saturating concentrations of c^{3+} (33). Fig. 3 shows that the model has no trouble fitting the data. In the original work, the authors argue that the pH dependence of the steady-state rate seen in Fig. 3 B was due to two pK_a values of 6.6 and 9.2 of the Rieske ISP based on a Michaelis-Menten type

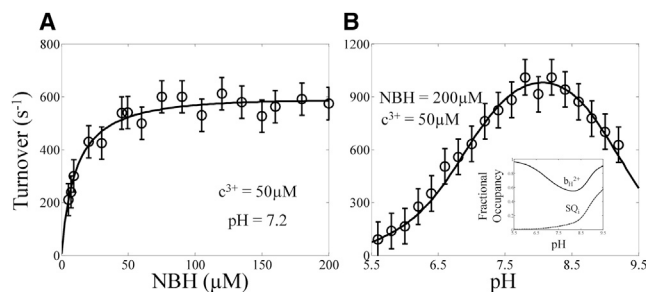


FIGURE 3 Turnover rate of reconstituted *bc*₁ complex in proteoliposomes oxidizing nonylubihydroquinone (NBH) (33). (A) The simulated turnover rate versus NBH concentration at fixed pH. (B) The simulated turnover rate versus pH at fixed c^{3+} and NBH concentrations. (Inset) Fractional occupancy of the reduced cyt *b*_H (solid) and stable SQ (dotted) as a function of pH. (Symbols) Experimental data. (Lines) Model simulations.

model. But cyclic voltammetry identified pK_a values of 7.6 and 9.2 (39). The *bc*₁ complex model presented here is able to explain this discrepancy and fit the data with the true pK_a values. This highlights the need for caution when interpreting data with simple models. This issue is discussed in more detail below.

The third and fourth data set were obtained under similar conditions where both consisted of isolated bovine *bc*₁ complex reduced with ubiquinol-2 to explore the effects of various substrate and product combinations. The primary difference between these two data sets is that the experimental buffer in the fourth data set contained 20 mM MgCl₂, which interferes with cyt *c* binding to the enzyme. The model captures the data very well across both data sets, as shown in Figs. 4 and 5.

In the fifth data set, energized isolated rat liver mitochondria supplemented with exogenous ubiquinol-2 and horse heart cyt *c* were used to investigate how the proton-motive force influenced the steady-state flux through the *bc*₁ complex. Rotenone and oligomycin were included in the respiration buffer, and the $\Delta\Psi$ was titrated with FCCP. The resultant steady-state oxygen consumption rates (equivalent to flux through the *bc*₁ complex) was measured in two respiration buffers designed to alter the proton-motive force via Δ pH generation (sucrose-based) and dissipation (KCl-based). The model is able to reproduce this data set very well as shown in Fig. 6 A. It should be noted that there exists an unknown interaction between exogenous ubiquinol-2 and

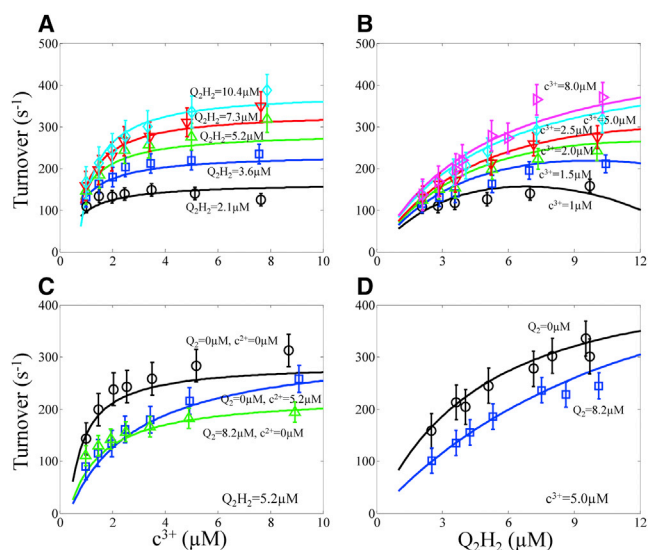


FIGURE 4 Turnover rate of isolated *bc*₁ complex oxidizing ubiquinol-2 (Q₂H₂) (16). (A) The simulated turnover rate versus c^{3+} concentration at varying Q₂H₂ concentrations. (B) The simulated turnover rate versus Q₂H₂ concentration at varying c^{3+} concentrations. (C) The simulated turnover rate versus c^{3+} concentration at varying Q₂ and c^{2+} concentrations at a fixed Q₂H₂ concentration. (D) The simulated turnover rate versus Q₂H₂ concentration at varying Q₂ concentrations at a fixed c^{3+} concentration. Fixed concentrations for each simulation are indicated on each panel. (Symbols) Experimental data. (Lines) Model simulations.

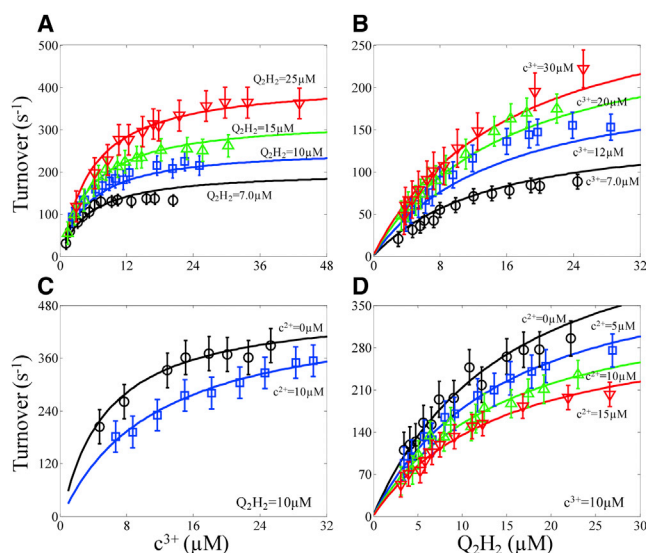


FIGURE 5 Turnover rate of isolated bc_1 complex oxidizing Q_2H_2 (18). (A) The simulated turnover rate versus c^{3+} concentration at various fixed Q_2H_2 concentrations. (B) The simulated turnover rate versus Q_2H_2 concentration at various fixed c^{3+} concentrations. (C) The simulated turnover rate versus c^{3+} concentration at various fixed c^{2+} concentrations at a fixed Q_2H_2 concentration. (D) The simulated turnover rate versus Q_2H_2 concentration at various fixed c^{2+} concentrations at a fixed c^{3+} concentration. (Symbols) Experimental data. (Lines) Model simulations.

endogenous ubiquinol-10. This is evident as shown in Table 2 by the different fitted ubiquinone binding constants when compared to the third and fourth data sets, which also used ubiquinol-2. Alternatively, the difference may be

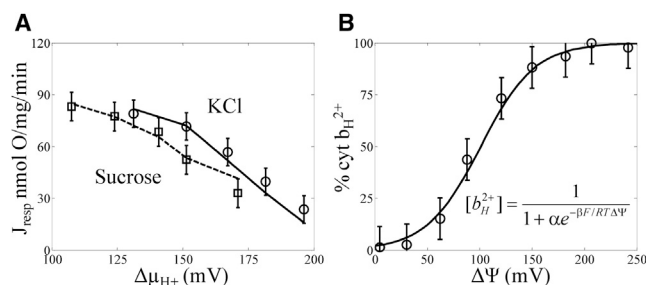


FIGURE 6 The effect of proton-motive potential on turnover (37) and extent of cyt b reduction (40). (A) The steady-state O_2 consumption rate of isolated rat liver mitochondria in the presence of rotenone and oligomycin versus the proton-motive force is shown. The experimental conditions impose that the rate of O_2 consumption is equivalent to twice the turnover rate for the bc_1 complex. The proton-motive force was titrated with FCCP. (Solid line) Simulated turnover rate when the mitochondria are suspended in a KCl-based respiration buffer. (Dotted line) Simulated turnover rate when the respiration buffer is switched to sucrose-based buffer. Exogenous Q_2H_2 and cytochrome c were supplied to the respiring mitochondria. The proton-motive force is defined as: $\Delta\mu_{H^+} = \Delta\Psi + 59\Delta pH$. (B) Reconstituted bc_1 complex was reduced with ascorbate and the extent of cyt b_H reduction was spectrophotometrically measured in the presence of varying K^+ diffusion potentials. The equation represents the fraction cyt b_H , reduced as a function of the $\Delta\Psi$. The parameters α and β were set to 50 and 0.5, respectively.

explained by the difference in microenvironments (detergent micelles versus mitochondrial membrane). Moreover, the effect of the exogenous cyt c on the endogenous cyt c pool is unclear. Despite these complications, this data set is particularly useful to identify the $\Delta\Psi$ -related parameters. To further corroborate the value of β given in Table 1, a supplementary data set was used (40). In this data set, reconstituted bovine bc_1 complex was reduced with ascorbate and the extent of cyt b_H reduction was spectrophotometrically measured in the presence of varying K^+ diffusion potentials as shown in Fig. 6 B. In the equation given in the inset, α was set to 50 and reflects the internal redox equilibrium values.

The protonated state of the Rieske ISP is believed to control the turnover rate of the bc_1 complex, but the precise mechanism is uncertain. The ISP contains two His residues with pKa values of 7.6 and 9.2 (39). This results in a precipitous drop in the midpoint potential as the pH is raised above 7. There is a consensus that rapid turnover requires at least one proton bound to the ISP, thus the enzyme is inhibited at alkaline pH. The model includes this type of regulation (see Eq. 11). However, the role of the protonated state of the His ligand with a pKa of 7.6, His¹⁶¹, is less certain. Previous work showed that the turnover rate exhibited a bell-shaped pH-dependence but at apparent pKa values of 6.6 and 9.2 (33). Targeted mutagenesis studies seem to support the argument that His¹⁶¹ participates in the formation of the ES-complex and thus controls turnover (41). This is even more confounding by a study that showed the protonated state of His¹⁶¹ had no influence over ubiquinol binding (42). In the model, it was assumed that the protonated state of His¹⁶¹ did not affect enzyme turnover, and the drop in catalytic activity for pH < 8 is due to the pH-dependent cyt b_H and stable SQ midpoint potentials. At acidic pH, state E1 is less populated because the electron at the Q_i -site is localized on cyt b_H as shown in Fig. 3 B inset (solid line). Thus, the transition to state E1 at low pH is inhibited. This is corroborated by redox titrations showing the stable SQ at the Q_i -site is more prevalent at alkaline pH (30,43), as shown in Fig. 3 B's inset (dotted line).

Recently, a redox-specific binding mechanism was proposed for the Q_i -site whereby ubiquinol preferentially binds to the oxidized enzyme and ubiquinone binds to the reduced enzyme (44). It was argued that this mechanism explained why an excess of ubiquinol over ubiquinone at the Q_i -site appeared to have a minimal influence over the steady-state turnover rates. Although the authors make convincing arguments in support of their hypothesis, the proposed mechanism leads to conditions that violate thermodynamics. At equilibrium, the product of state transition rates around a closed loop should equal the product in the opposite direction (45,46). However, with the mechanism described above, the following imbalanced expression is obtained: $P_{Q_i,red} = P_{Q_i,ox}$, where $P_{Q_i,red}$ and $P_{Q_i,ox}$ are the binding polynomial for the reduced and oxidized enzyme at the Q_i -site, respectively (see proof in the Supporting Material).

In other words, the ubiquinone binding polynomial for the Q_i-site must be the same regardless of the redox state of the enzyme. Analysis of the model presented here suggests that this phenomenon is best explained by the high intrinsic rate constants for the Q_i-site related state transitions. This leads to relatively little inhibition at the Q_i-site by ubiquinol via a mass-action mechanism, especially in the absence of a proton-motive force.

The native ubiquinone substrate for the *bc*₁ complex is ubiquinol-10. But its partition coefficient is $>10^{20}$ (47), so it is impractical to explore the enzyme kinetics of isolated or reconstituted mammalian *bc*₁ complex with its native substrate. Therefore, kinetic assays are typically done with Q-analog substrates that are less hydrophobic. Nearly all of these Q-analogs differ structurally from the native substrate by having a shorter isoprenoid tail or a different carbon-based tail. In its native environment, the hydrophobic tail of ubiquinol-10 causes the molecule to partition more frequently in the midplane of the membrane (48,49). Therefore, ubiquinol-10 may have a kinetic advantage over the Q-analogs by possessing an ability to easily traverse the Q_o-site and Q_i-site entry portals. This makes a detailed analysis of the fitted ubiquinone binding constants less meaningful when the objective is to simulate the *bc*₁ complex with its native substrates. Although the fitted ubiquinone binding constants presented in Table 2 closely align with the reported values, they still do not reflect actual binding constants and must be interpreted with due diligence (for details, see discussion in Crofts et al. (50)). Moreover, the Q-analogs have varying partition coefficients in the hydrophobic environments for both their oxidized and reduced forms (16,47). Therefore, the true Q-analog concentration within the hydrophobic environment cannot be ascertained with certainty. To circumvent this issue, more physiological ubiquinone binding constants, specific for ubiquinol-10, were sought. An extensive sensitivity and control analysis of the model using the parameters presented in Table 3 is given in the Supporting Material. These more relevant parameter values were used for the superoxide simulations discussed below.

The superoxide simulations shown in Fig. 7 were performed with the fixed parameters in Table 1, the fitted state transition-rate constants in Table 2, and parameters in Table 3. The parameters in Table 3 were obtained from estimates given in the literature and tuned so that the model simulations produced physiologically relevant turnover and superoxide production rates. The cyt *c* binding constants were assumed to be similar to the fitted constants for horse heart cyt *c*, based on its apparent universal nature as a substrate for *bc*₁ complexes from other species (51). The Q_o-site binding constants were set according to reports that indicate the ubiquinone-10 binding constant at the Q_o-site is anywhere between 0.5 and 5 mM (28,52) and that ubiquinol-10 binds with near equal affinity (53). The ubiquinone-10 binding constant at the Q_i-site was tuned to give physiological

TABLE 3 Native substrate model parameters

Parameter	Definition	Value	Reference
K_{c3}	Cytochrome c^{3+} binding constant	1.1 μ M	— ^a
K_{c2}	Cytochrome c^{2+} binding constant	1.2 μ M	— ^a
$K_{QH_2}^o$	Q ₁₀ H ₂ Q _o -site binding constant	1.0 mM	(1) ^b
K_{Q}^o	Q ₁₀ Q _o -site binding constant	0.8 mM	(1,52,53) ^c
K_{Q}^i	Q ₁₀ H ₂ Q _i -site binding constant	1.0 mM	(54) ^d
$K_{Q_2}^i$	Q ₁₀ Q _i -site binding constant	2.2 mM	— ^e
$[bc_1]$	Mitochondrial <i>bc</i> ₁ complex content	80 pmol/mg	(64)
$[c]_{tot}$	Total mitochondrial cytochrome <i>c</i> concentration	200 μ M	(65) ^f
$[Q]_{tot}$	Total mitochondrial ubiquinone concentration	20 mM	(64) ^g
$[O_2]$	Oxygen concentration in respiration buffer	200 μ M	— ^h
$[O_2^{\cdot-}]$	Superoxide concentration in respiration buffer	0.1 μ M	(22)

^aFitted value in Table 2.

^bBased on the observation that maximum activity in the absence of a proton-motive force when the Q-pool is 50% reduced.

^cIn the range given, and the Q_o-site affinity for QH₂ and Q are very similar.

^dBased on state-2 respiration rates of 30 nmol O₂/mg/min when $\Delta\Psi = 200$ mV and $\Delta pH = 0$.

^eDefined via microscopic reversibility where $K_{QH_2}^i = K_{QH_2}^o K_{Q_2}^i / K_{Q_2}^o$.

^fAssuming 1 μ L/mg IMS volume for isolated mitochondria.

^gAssuming 0.2 μ L/mg inner mitochondrial membrane lipid space volume.

^hThe amount of dissolved oxygen in standard respiration buffer based on atmospheric conditions at room temperature.

steady-state rates for state-2 respiratory conditions. State-2 respiration is characterized by a low oxygen consumption rate of ~ 30 nmol O₂/mg/min and associated with a high proton-motive force near 200 mV (54). The ubiquinol-10 binding constant at the Q_i-site is a function of the other binding constants and determined via microscopic reversibility.

Fig. 7 A shows the surface plot of the simulated steady-state rate of ubiquinol oxidation as a function of $\Delta\Psi$ and the percent that the Q-pool is reduced. The cyt *c*-pool was set to 90%, reduced to reflect physiological conditions (54,55). At low proton-motive force, the rate reaches a maximum when the Q-pool is 50% reduced as reported in Crofts (1). The corresponding fractional occupancies of the state model are shown in Fig. 7 B. Fig. 7 C shows the simulated surface plot of superoxide production for the same conditions as in Fig. 7 A. As the proton-motive force increases and the Q-pool becomes more reduced, the superoxide production rate dramatically increases. State E4 comprises the largest fraction except when there is a large proton-motive force or the Q-pool is highly oxidized.

The precise contribution of superoxide generation by the *bc*₁ complex is uncertain. As such, the intrinsic state transition-rate constants for the auxiliary states were tuned to recapitulate the contemporary consensus of reactive oxygen species (ROS) generation via the respiratory chain. Model

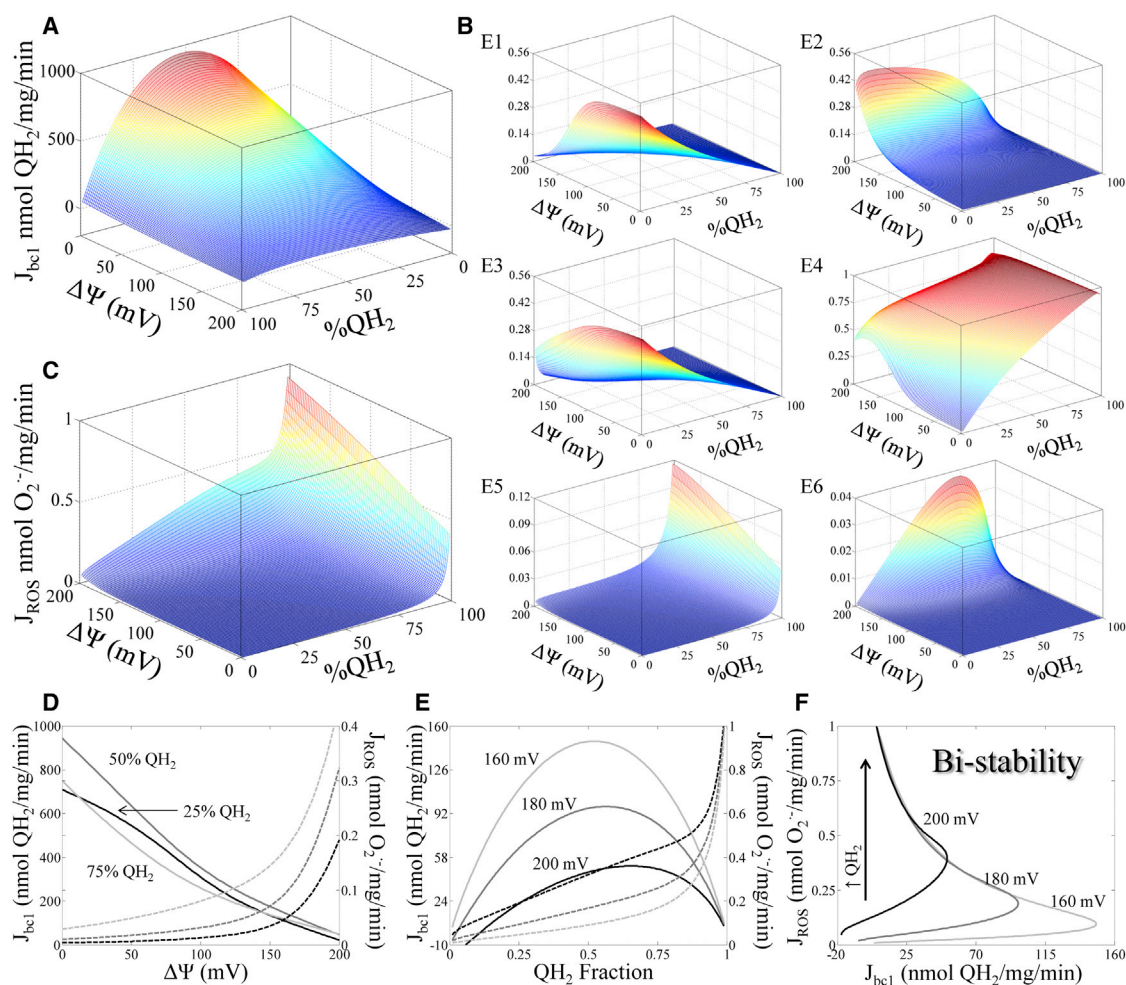


FIGURE 7 Model simulations with the native substrate parameters of flux through the *bc1* complex, superoxide production, and the bistability phenomenon. (A) The surface plot of the rate of QH₂ oxidation simulated with the model as a function of $\Delta\Psi$ and the percent the Q-pool is reduced is shown. (B) The surface plots for the fractional state occupancies are shown for the same conditions in panel A. (C) The corresponding surface plot of the rate of superoxide production under the same conditions in panel A is shown. In all simulations the cytochrome *c* pool was 90% oxidized, and the pH at the P-side and N-side was 7. (D) The rate of QH₂ oxidation (solid line, left y axis) and the rate of superoxide production (dotted line, right y axis) are shown as a function of $\Delta\Psi$ at three different reduced Q-pool levels. (E) The rate of QH₂ oxidation (solid line, left y axis) and the rate of superoxide production (dotted line, right y axis) is shown as a function of the reduced Q-pool level at three different $\Delta\Psi$ values. (F) A phase plot of the rate of superoxide production versus QH₂ oxidation is shown at three different $\Delta\Psi$ values to demonstrate the bistability phenomenon. (Arrow) Direction of increasingly reduced Q-pool levels. In all simulations the cytochrome *c* pool was 90% oxidized, and the pH at the P-side and N-side was 7.

simulation results shown in Fig. 7, D and E (various composite slices through Fig. 7, A and C, respectively), demonstrate a qualitative match to the available data on ROS generation. Fig. 7 D shows that as the $\Delta\Psi$ increases, the steady-state rate of ubiquinol oxidation decreases as the superoxide production rate exponentially increases. This simulated phenomenon matches a qualitative experimental data set describing superoxide production rate versus $\Delta\Psi$ (2). Likewise, as the Q-pool is reduced, the rate of superoxide production increases as shown in Fig. 7 E. The simulated superoxide production rates are in the experimental range given by recent experimental ROS emission data when the ROS scavenging system is inhibited (56). Specifically, the superoxide production rate is ~1% of the *bc1* complex flux under state-2 respiratory conditions (30–60 nmol

QH₂/mg/min and 200 mV). The model is also able to simulate significantly increased levels of superoxide production under saturating antimycin A conditions (not shown). Fig. 7 F shows that the model is able to reproduce the bistability phenomenon whereby different rates of superoxide production are maintained when the enzyme is differentially reduced whereas the overall flux through the enzyme is the same. For example, at a flux rate of 30 nmol QH₂/mg/min when the $\Delta\Psi$ equals 180 mV, there are two possible rates of superoxide production. One of the rates is very low and produces only 0.05 nmol O₂⁻/mg/min, but when the environment is more reduced, the rate that superoxide is produced dramatically increases an order of magnitude to 0.5 nmol O₂⁻/mg/min. This switch from low to high superoxide production rates is the bistability phenomenon and

believed to be partly responsible for ischemia/reperfusion injury (24,27). Targeted therapies aimed at preventing this mode of superoxide production would prove most beneficial during ischemia/reperfusion.

Several groups have proposed that the *bc*₁ complex operates as a functional dimer characterized by rapid intermonomer electron transfer (57–59). If true, this hypothesis would require a reevaluation of the Q-cycle and alter contemporary understanding of the *bc*₁ complex's catalytic mechanism. In contrast, the work presented here demonstrates that such a mechanism is unnecessary to explain the experimental data used to parameterize the model (16,18,19,33,37). In fact, more recent work reveals that the functional dimer results obtained thus far may be incorrectly interpreted (60). If the *bc*₁ complex is unambiguously shown to operate as a functional dimer, the model can be extended to include additional states to explore this phenomenon in the future.

CONCLUSIONS

Presented here is a thermodynamically consistent kinetic model of *bc*₁ complex in mammalian mitochondria. It consists of six distinct states characterized by the electron distribution in the enzyme where each state is made up of a number of substates that correspond to various electron localizations within each state. The model incorporates all major redox centers near the Q_o- and Q_i-site of the enzyme, includes all the pH-dependent redox reactions, accounts for the effect of the proton-motive force on the reaction rate, simulates superoxide production at the Q_o-site, and reproduces the bistability phenomenon. Integrating this model into existing mitochondrial respiration models would produce more realistic model simulations and help uncover the intricate relationship between superoxide production and mitochondrial bioenergetics.

SUPPORTING MATERIAL

Supporting analysis, two figures, and references (66–68) are available at [http://www.biophysj.org/biophysj/supplemental/S0006-3495\(13\)00616-4](http://www.biophysj.org/biophysj/supplemental/S0006-3495(13)00616-4).

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Supporting Material A

Analysis of the kinetics and bistability of ubiquinol:cytochrome c oxidoreductase

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Description of state transition-rates

$$k_{12} = k_{120} \frac{[QH_2]}{K_{QH_2}^o P_{Qo}} \frac{[c^{3+}]}{K_{c3} P_c} \frac{[H^+]_p (1 + [H^+]_p / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} + k_{SO} e^{\Delta G_{SO}/RT} \frac{[O_2^-][Q]}{K_Q^o P_{Qo}} \quad (S1A)$$

The transition-rates were derived based on the law of mass action. As an example, the derivation of k_{12} is presented. For convenience, the equation is given above. For the transition from E1 to E2, the bc₁ complex must be bound with ubiquinol and cytochrome c³⁺ and the ISP must be in the correct protonated state. The term, $[QH_2] / K_{QH_2}^o / P_{Qo}$, is the fraction of E1 bound with ubiquinol. Similarly, the term, $[c^{3+}] / K_{c3} / P_c$, is the fraction of E1 bound with cytochrome c³⁺. The ISP protonated state is characterized by the term, $[H^+]_p (1 + [H^+]_p / 10^{-pK_{ISPox1}}) / 10^{-pK_{ISPox2}} / P_{ISP}$. Thus, the fraction of E1 that transitions to E2 is the product of these terms times the State E1 to E2 intrinsic transition-rate constant, k_{120} . The superoxide term on the far right-hand side is the result of state E2 transitioning to E1 via superoxide oxidation, a rare occurrence considering that ΔG_{SO} at pH 7 equals approximately -4.7 kJ/mol. All the transition rates presented herein were derived in a similar manner.

Experimental data overview

The model was parameterized using experimental data from the literature. The fixed parameters are located in Table 1. The adjustable parameters shown in Table 2 were fitted to five independent data sets. Each data set used a Q-analog to assay the kinetics of the bc₁ complex. Four of the five data sets characterized the kinetics of isolated or reconstituted bovine bc₁ complex including the initial rate substrate-dependence, pH-dependence and end-product inhibition (1-4). The fifth data set was obtained using isolated rat liver mitochondria (5). This data set shows how the steady-state respiration rate (equivalent to flux through the bc₁ complex) is controlled by exogenously added ubiquinol-2 and cyt c as a function of the $\Delta\Psi$ and pH gradient across the inner mitochondrial membrane. During the parameter estimation, the relevant kinetic parameters were identical for the respective simulations when the type of substrates used and experimental conditions were alike.

In all experiments, the Q-analogs exhibited non-enzymatic oxidation via interaction with cyt c that significantly increases with pH (6, 7). This was usually monitored for 1-3 minutes before initiating the enzymatic reaction and subtracted from the measured rate of cyt c reduction. This rate was reported to be near 1% of the enzymatic rate (3), not reported but measured (1, 4) or not even measured (2). The details given in the experimental methods do not allow this phenomenon to be accurately modeled; therefore, an approach to estimate it using an additional adjustable parameter called the Q_{ratio} was favored. The Q_{ratio} is a measure of how much cyt c was produced from non-enzymatic oxidation of the Q-analog before the enzymatic reaction was initiated. This value was used to correct the reported substrate and product concentrations. For example, Table 2 shows that the Q_{ratio} for Data Set 1 is 0.33%. This means that 0.33% of the initial ubiquinol was oxidized to ubiquinone and reduced twice as much cyt c before the enzyme catalyzed reaction was started.

Fitting and sensitivity analysis

The model was fit to the data using Eq. S2A:

$$f = \sum_k \frac{1}{M_k} \sum_{j=1}^{M_k} \frac{1}{N_{j,k}} \sum_{i=1}^{N_{j,k}} \left(\frac{y_{i,j,k}(p) - Y_{i,j,k}}{\sigma_{i,j,k}} \right)^2 \quad (\text{S2A})$$

where f is the objective function evaluated at a parameter point p , $y_{i,j,k}$, $Y_{i,j,k}$ and $\sigma_{i,j,k}$ are the model output, experimental data and measurement uncertainty (standard deviation), respectively, corresponding to the i th experimental data point in the j th experimental data curve for the k th data set. $N_{j,k}$ is the number of data points in the j th experimental data curve for the k th data set and M_k is the number of data curves for the k th data set. When no measurement uncertainty was given with the experimental data, it was assumed to be 10% of the maximum value for a given data curve.

The normalized sensitivity coefficients were computed using Eq. S3A:

$$S_{lm} = \frac{\partial y_l}{\partial p_m} \frac{p_m^*}{y_l} \quad (\text{S3A})$$

where the normalized model output y_l consists of the concatenation of the i th, j th and k th simulated output at every sampling point for all data curves and data sets, m is the index of the parameters and p^* is the best identified parameter value in terms of maximum likelihood. The model output derivatives with respect to the parameters were estimated using the complex method (8). The sensitivity values given in Table 2 are the computed averages using all the non-zero elements in the matrix S_{lm} for the m th parameter.

Supporting Material B

This section demonstrates that a redox-specific binding mechanism for the Q_i-site where ubiquinol preferentially binds to the oxidized enzyme and ubiquinone binds to the reduced enzyme violates thermodynamics. Without loss of generality, we assume the absence of superoxide production.

Definitions

This mechanism would require that the binding polynomial for the oxidized Q_i-site, P_{Qi}^o , favor binding of QH₂, and the binding polynomial for the reduced Q_i-site, P_{Qi}^r , favor binding of Q. Thus, P_{Qi}^o must not equal P_{Qi}^r . Demonstration of the infeasibility of this mechanism uses the equilibrium constant

$$K_{eq} = \frac{[Q][c^{2+}]^2}{[QH_2][c^{3+}]^2} \Big|_{eq} = K_o^2 K_i^2 K_{Qo}^2 K_{Qi,1} K_{Qi,2} e^{-2\beta F \Delta \Psi / RT}, \quad (S1B)$$

which defines the overall equilibrium of the reaction of Eq. 1. The quantities K_{Qo} , $K_{Qi,1}$ and $K_{Qi,2}$ are defined in Eqs. 13, 17 and 21. We introduce two new equilibrium constants K_o and K_i

$$K_o = e^{-\Delta G_o / RT}, \quad (S2B)$$

and

$$K_i = e^{-\Delta G_i / RT}, \quad (S3B)$$

where the free energies ΔG_o and ΔG_i are defined in Eqs. 24 and 25. Equations S2B and S3B define the equilibrium for the first electron oxidation of ubiquinol and the electron transfer from cyt b_L to cyt b_H, respectively.

The product of the equilibrium constants on the far right hand side of Eq. S1B is equivalent to

$$K_{Qo}^2 K_{Qi,1} K_{Qi,2} K_o^2 K_i^2 e^{-2\beta F \Delta \Psi / RT} = e^{-(2(\Delta G_o + \Delta G_i + \Delta G_{Qo}) + \Delta G_{Qi,1} + \Delta G_{Qi,2}) / RT - 2\beta F \Delta \Psi / RT}. \quad (S4B)$$

From Eqs. 12, 16, 20, 24 and 25, the exponent terms in Eq. S4B are rewritten in terms the midpoint potentials as

$$\begin{aligned} 2(\Delta G_o + \Delta G_i + \Delta G_{Qo}) + \Delta G_{Qi,1} + \Delta G_{Qi,2} = & -F(2(E_m(b_L^{3+} / b_L^{2+}) - E_m(Q / SQ_o)) + (E_m(c^{3+} / c^{2+}) \\ & - E_m(SQ_o / QH_2) + E_m(b_H^{3+} / b_H^{2+}) - E_m(b_L^{3+} / b_L^{2+})) + E_m(Q / SQ_i) - E_m(b_H^{3+} / b_H^{2+}) + E_m(SQ_i / QH_2) \\ & - E_m(b_H^{3+} / b_H^{2+}) - 2\Delta \Psi). \end{aligned} \quad (S5B)$$

The β term in the exponent in Eq. S4B cancels with the β from Eq. 20 to leave $-2\Delta \Psi$ in Eq. S5B. This equation reduces to

$$2(\Delta G_o + \Delta G_i + \Delta G_{Qo}) + \Delta G_{Qi,1} + \Delta G_{Qi,2} = -2F(E_m(c^{3+} / c^{2+}) - E_m(Q / QH_2) - \Delta \Psi), \quad (S6B)$$

which equals the ΔG for the overall reaction given in Eq. 1 at a specified pH and $\Delta \Psi$. The pH contribution to the free energy of reaction is included in the ubiquinol midpoint potentials as shown in Eqs. 3 and 7.

Proof

At equilibrium, the product of state transition-rates around a closed loop must equal the product in the opposite direction. For the primary states, we have

$$k_{12} k_{23} k_{34} k_{41} = k_{14} k_{43} k_{32} k_{21}. \quad (S7B)$$

In the absence of superoxide production and substituting the definitions for the state transition-rates, Eq. S7B is written as

$$\begin{aligned}
& k_{120} \frac{[QH_2]_{eq}}{K_{QH_2}^o P_{Qo}} \frac{[c^{3+}]_{eq}}{K_{c3} P_c} \frac{[H^+]_p (1 + [H^+]_p / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} \times k_{230} \frac{r_{Qo}}{f_{Qo}} \frac{r_{Qi,2}}{f_{Qi,2}} e^{-\beta F \Delta \Psi / 2 RT} \\
& \times k_{340} \frac{[QH_2]_{eq}}{K_{QH_2}^o P_{Qo}} \frac{[c^{3+}]_{eq}}{K_{c3} P_c} \frac{[H^+]_p (1 + [H^+]_p / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} \times k_{410} \frac{r_{Qo}}{f_{Qo}} \frac{r_{Qi,1}}{f_{Qi,1}} e^{-\beta F \Delta \Psi / 2 RT} \\
& = k_{410} e^{\Delta G_i / RT} \frac{e^{\beta F \Delta \Psi / 2 RT}}{f_{Qi,2}} \times k_{340} e^{\Delta G_o / RT} \frac{[c^{2+}]_{eq}}{K_{c2} P_c} \frac{1}{f_{Qo}} \frac{[H^+]_p (1 + [H^+]_p / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}} \\
& \times k_{230} e^{\Delta G_i / RT} \frac{e^{\beta F \Delta \Psi / 2 RT}}{f_{Qi,1}} \times k_{120} e^{\Delta G_o / RT} \frac{[c^{2+}]_{eq}}{K_{c2} P_c} \frac{1}{f_{Qo}} \frac{[H^+]_p (1 + [H^+]_p / 10^{-pK_{ISPox1}})}{10^{-pK_{ISPox2}} P_{ISP}}.
\end{aligned} \tag{S8B}$$

Canceling out terms on both sides; replacing the r terms by their respective definitions given in Eqs. 14, 18 and 22; moving all concentration terms to the right-hand side and other terms to the left-hand side, and grouping like terms, Eq. S8B reduces down to the following expression

$$K_o^2 K_i^2 K_{Qo}^2 K_{Qi,1} K_{Qi,2} e^{-2\beta F \Delta \Psi / RT} \times \frac{K_Q^{o2} K_{QH_2}^i K_{c2}^2}{K_{QH_2}^o K_Q^i K_{c3}^2} \times \frac{P_{Qo}^2 P_c^2 P_{Qi}^o}{P_{Qo}^2 P_c^2 P_{Qi}^r} = \frac{[Q][c^{2+}]^2}{[QH_2][c^{3+}]^2} \Bigg|_{eq}, \tag{S9B}$$

where P_{Qi}^o and P_{Qi}^r are the binding polynomials for the Qi-site when oxidized or reduced, respectively.

From the definition of K_{eq} given in Eq. S1B, Eq. S9B reduces to

$$\frac{K_Q^{o2} K_{QH_2}^i K_{c2}^2}{K_{QH_2}^o K_Q^i K_{c3}^2} \times \frac{P_{Qo}^2 P_c^2 P_{Qi}^o}{P_{Qo}^2 P_c^2 P_{Qi}^r} = 1. \tag{S10B}$$

As a result of the microscopic reversibility constraint, the kinetic constant term on the far left-hand side must equal 1.

Thus, Eq. S10B becomes

$$\frac{P_{Qo}^2 P_c^2 P_{Qi}^o}{P_{Qo}^2 P_c^2 P_{Qi}^r} = 1. \tag{S11B}$$

After canceling binding polynomials that appear in both the numerator and denominator, Eq. S11B becomes

$$\frac{P_{Qi}^o}{P_{Qi}^r} = 1. \tag{S12B}$$

Equation S12B requires that P_{Qi}^o equal P_{Qi}^r ; therefore, the redox-specific binding mechanism for the Qi-site is not thermodynamically feasible.

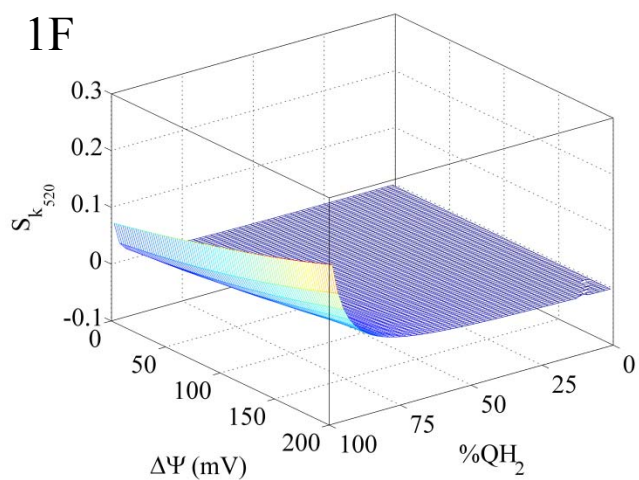
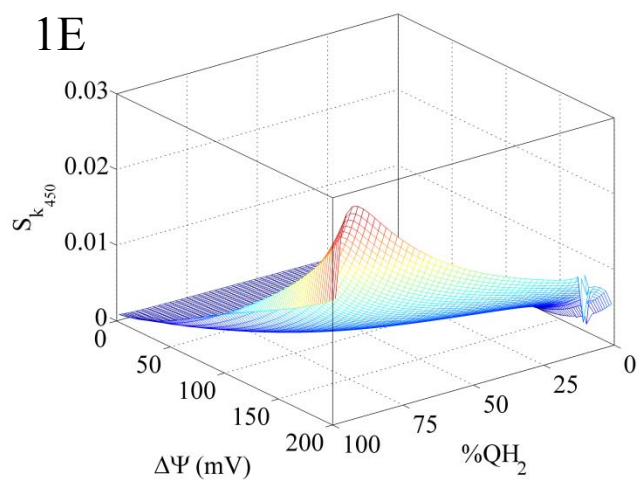
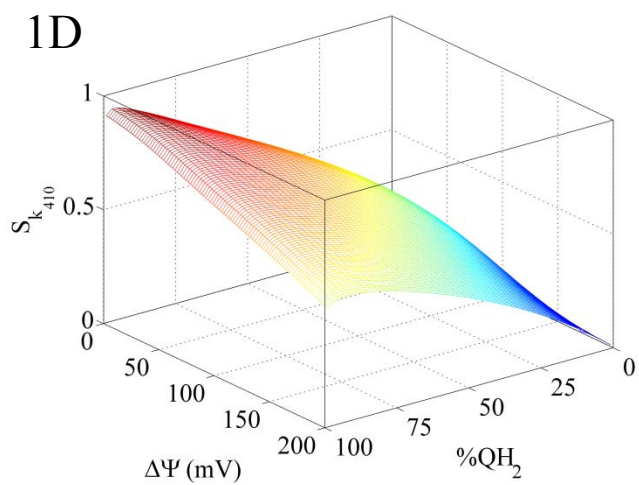
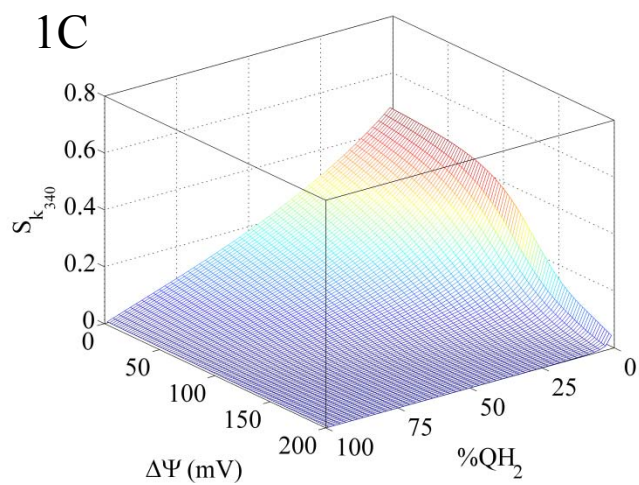
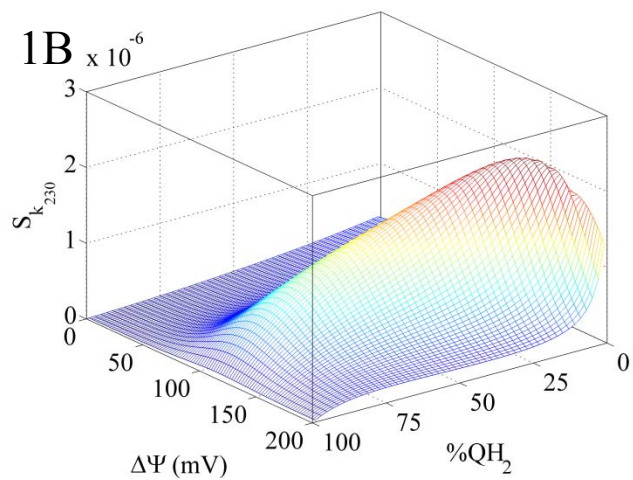
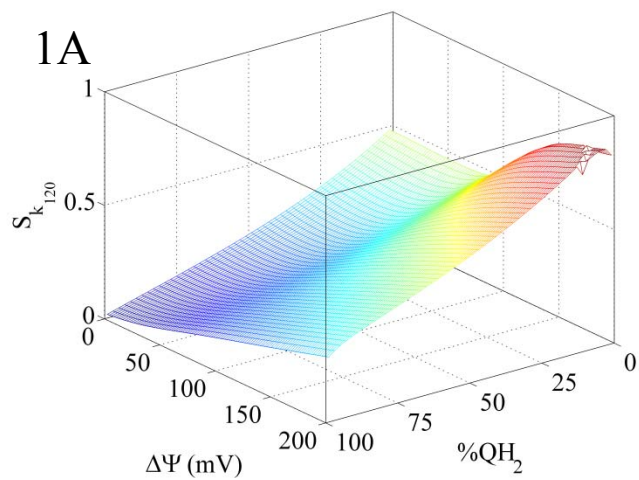
Supporting Material C

This section reports computed normalized sensitivity coefficients for the steady-state rate of cytochrome c reduction (Figs. 1A-1N) and superoxide production (Figs. 2A-2N) given in Eqs. 43 and 45, respectively. The conditions employed are the same as those for the J_{bc1} and J_{ROS} landscapes presented in Figure 7. The sensitivities were computed using Eq. S3A. For J_{bc1} , they are analogous to flux control coefficients. As such, the normalized sensitivity coefficients for J_{bc1} of the state transition-rate constants sum to unity (not shown). The flux control summation theorem is not applicable for the normalized sensitivity coefficients for J_{ROS} . The parameters chosen to examine were the following: k_{120} , k_{230} , k_{340} , k_{410} , k_{450} , k_{520} , k_{260} , k_{640} , K_{c3} , K_{c2} , $K_{QH_2}^o$, K_Q^o , K_Q^i and k_{SO} . The parameter $K_{QH_2}^i$ is constrained via microscopic reversibility as discussed above and in Table 3. Any non-smoothness in the surface plots is a result of numerical artifacts obtained during gradient approximation (e.g. see right corner in 1N). A brief summary of the results is given below.

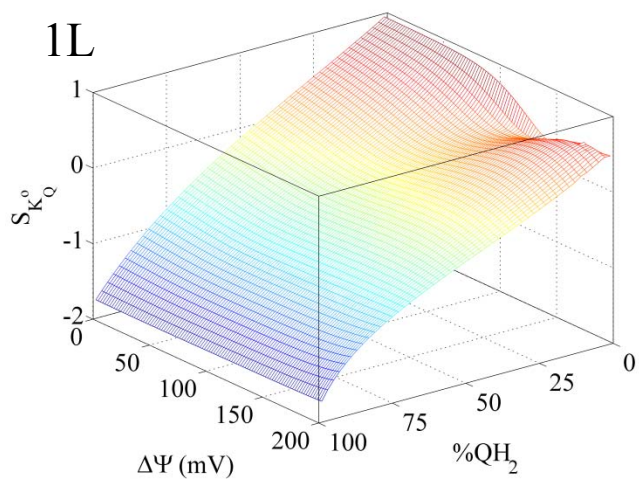
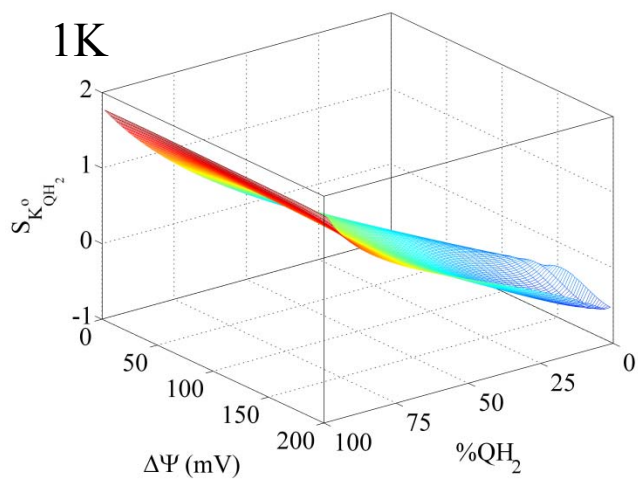
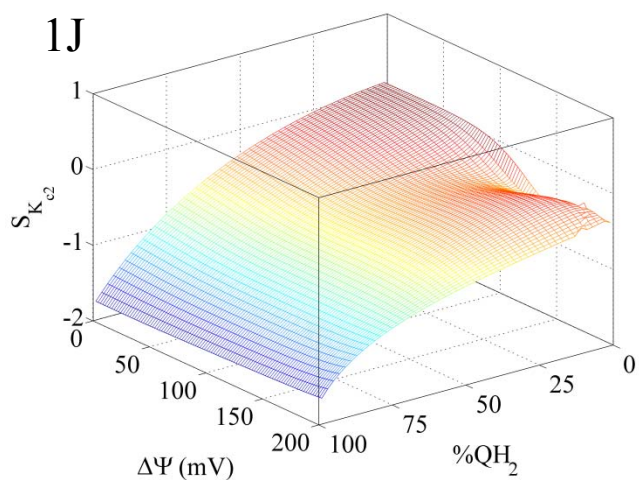
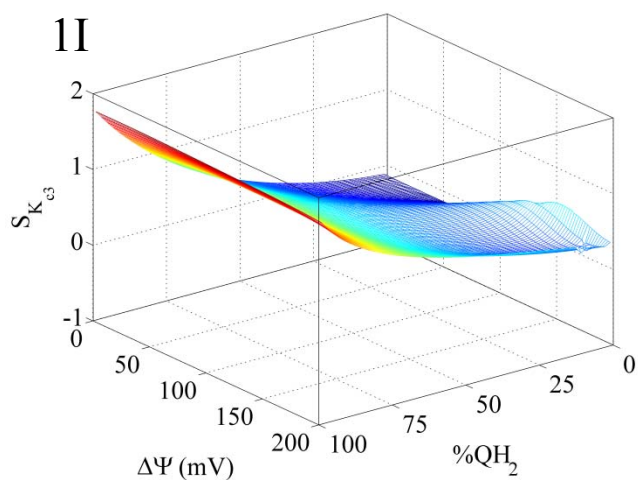
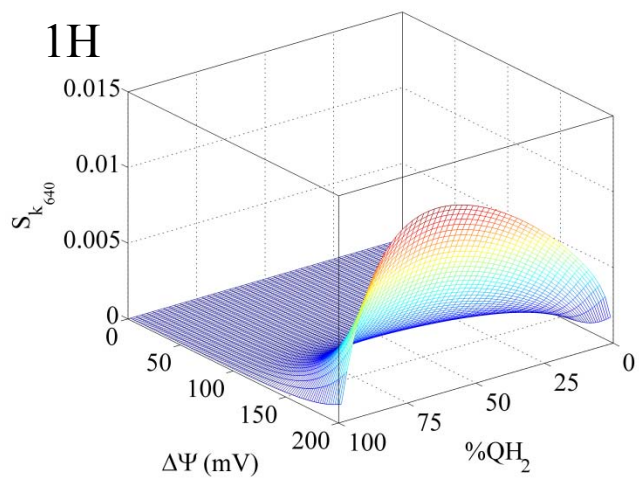
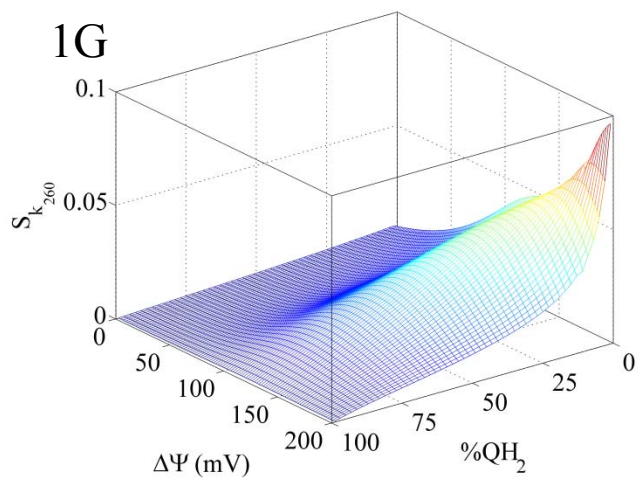
The normalized sensitivity coefficients for parameters listed above demonstrate how the control of the steady-state rate of cytochrome c reduction (Figs. 1A-1N) is distributed. The results reinforce the parameter sensitivities summarized in Table 2. However, the results shown below are for more physiologically relevant conditions that are outlined in the discussion for the native substrate parameter simulations. In general, k_{120} is controlling when $\Delta\Psi$ is high and the Q-pool is oxidized; k_{230} is not very important; k_{340} and k_{120} are equally controlling when $\Delta\Psi$ is low and the Q-pool is oxidized; the control by k_{410} increases as $\Delta\Psi$ decreases and the Q-pool becomes reduced; and the auxiliary state transition-rate constants have a minimal impact on the steady-state turnover of the bc₁ complex. The kinetic constant normalized sensitivity coefficients show that the substrate/product pairs exert opposite effects on J_{bc1} ; however, a more detailed analysis reveals a counter-intuitive effect. For example, an increase in K_{c3} should lead to a reduction in J_{bc1} since less c^{3+} is bound to the complex, but the normalized sensitivity coefficient is positive for nearly all $\Delta\Psi$ and %QH₂ combinations. This effect is due to the microscopic reversibility constraint discussed above. As a result of this constraint, $K_{QH_2}^i$ is also increased which results in more Q binding to the Q_i-site and slightly increasing J_{bc1} . It follows that only when the Q-pool is significantly oxidized does an increase in K_{c3} result in a decrease in J_{bc1} . Lastly, as expected, the superoxide production rate constant, k_{SO} , has minimal effect on J_{bc1} .

The normalized sensitivity coefficients for the superoxide production rate are shown in Figs. 2A-2N. When $\Delta\Psi$ is low and the Q-pool is oxidized, an increase in either k_{120} or k_{340} lead to an increase in J_{ROS} by increasing the fractional occupancies of the auxiliary states. This occurs via the E2 to E6 and E4 to E5 state transitions, respectively. As with J_{bc1} , k_{230} is not important. The parameter k_{120} becomes important as $\Delta\Psi$ decreases and the Q-pool becomes oxidized. Interestingly, an increase in k_{520} decreases J_{ROS} by lowering the steady-state levels of E5 and hence SQ₆. This is particularly important as the Q-pool is more reduced. The kinetic constants have opposite effects from their respective substrate/product pairs in the same manner with J_{bc1} . For example, increasing K_{c3} leads to a decrease in J_{ROS} by decreasing the fractional occupancy of states E2 and E4. These states lead to the superoxide producing states E5 and E6. Finally, k_{SO} is extremely important for J_{ROS} , as expected.

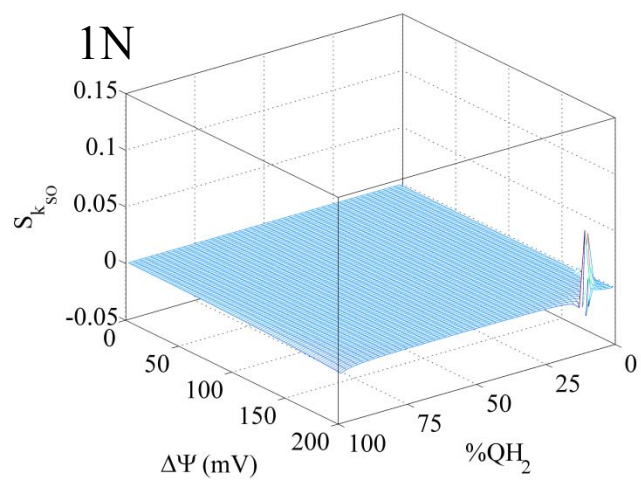
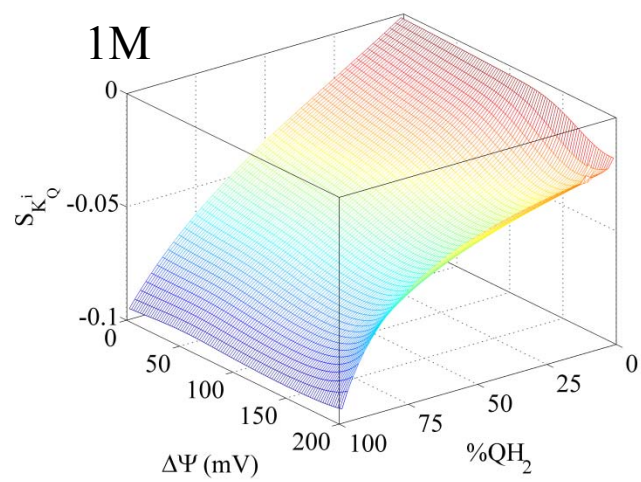
Normalized Sensitivity Coefficients for J_{bcI}



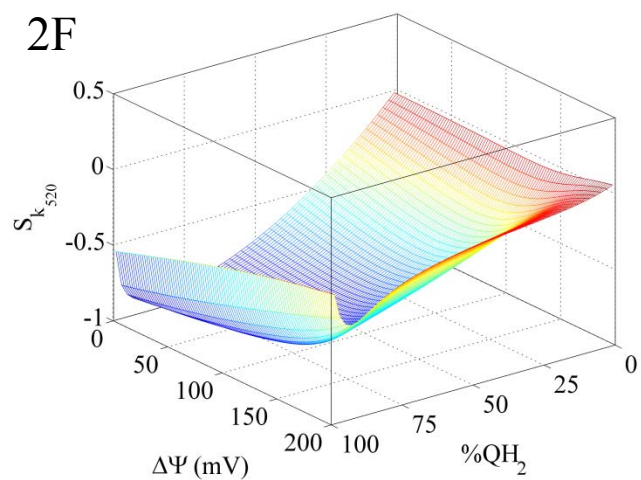
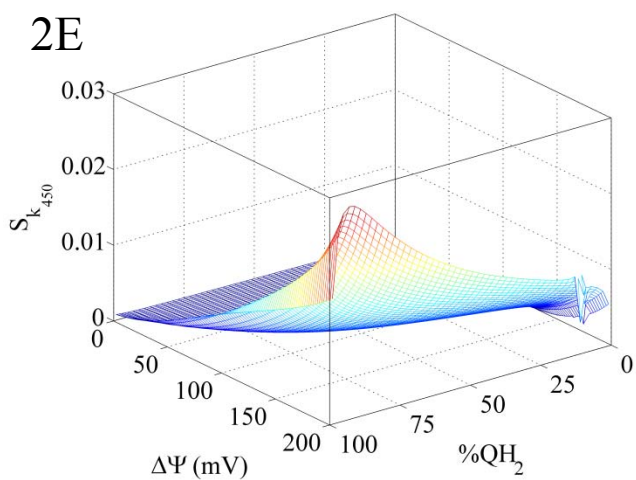
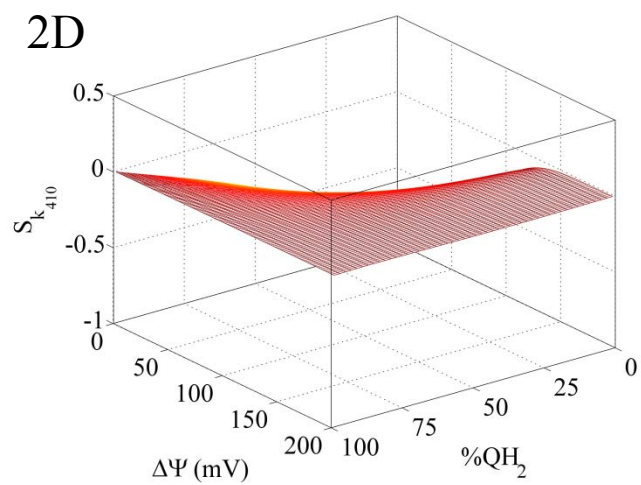
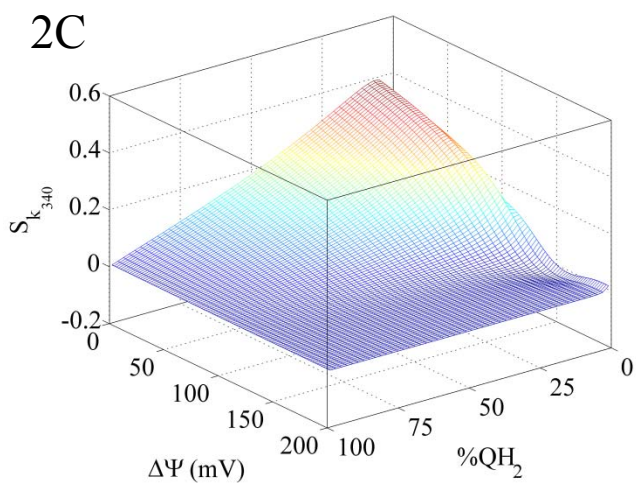
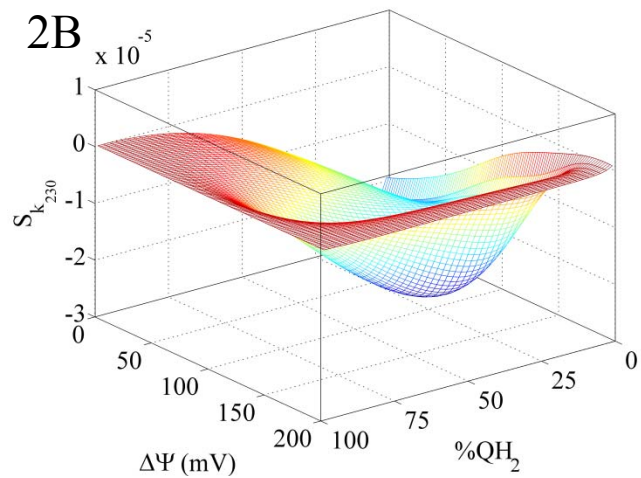
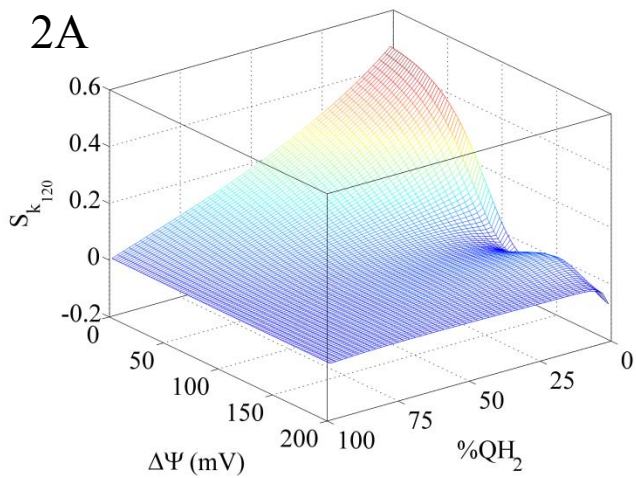
Normalized Sensitivity Coefficients for J_{bcI}



Normalized Sensitivity Coefficients for J_{bcI}

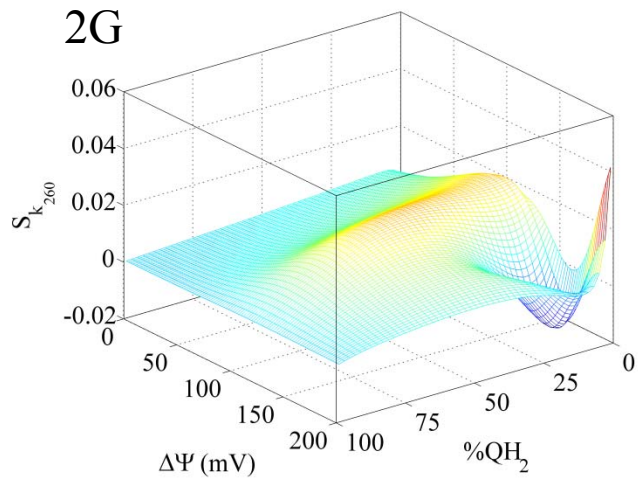


Normalized Sensitivity Coefficients for J_{ROS}

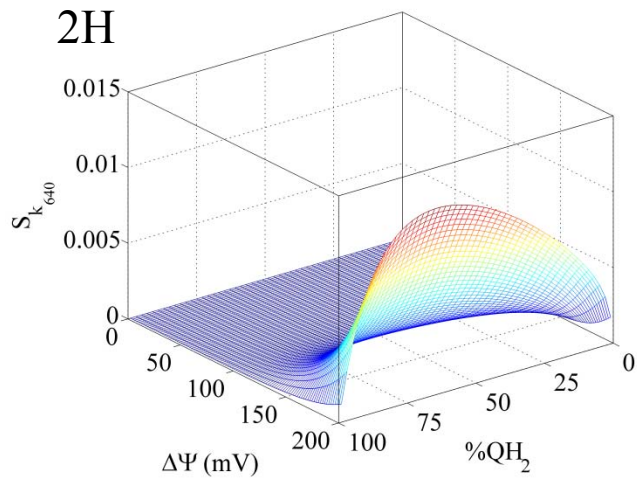


Normalized Sensitivity Coefficients for J_{ROS}

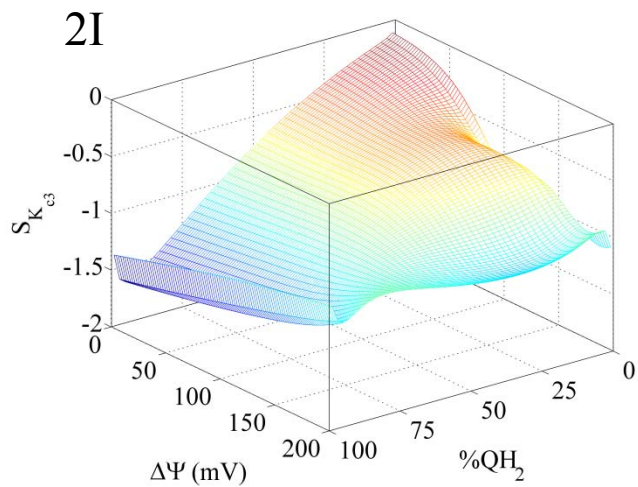
2G



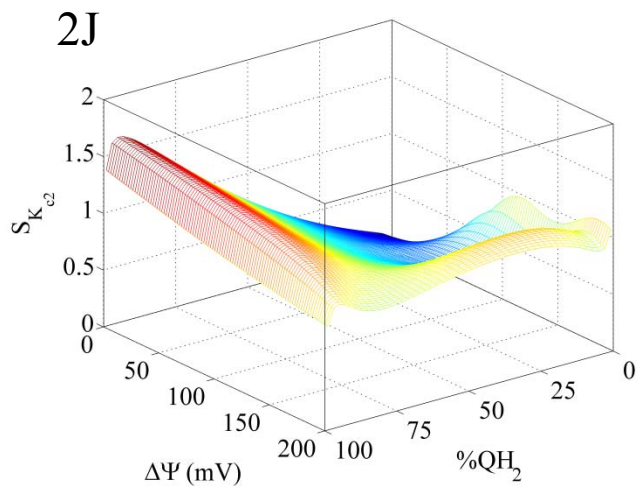
2H



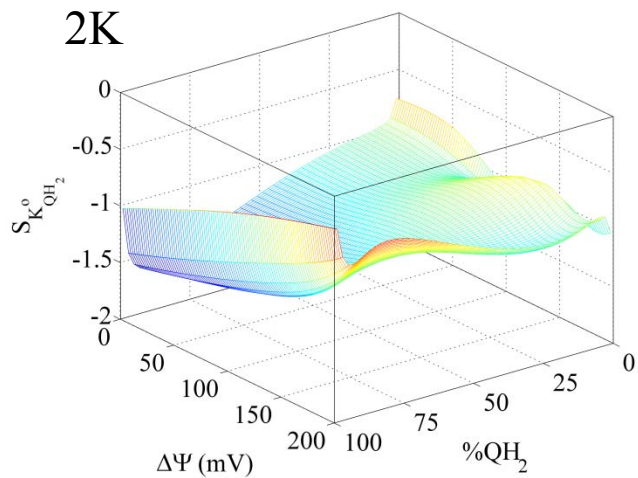
2I



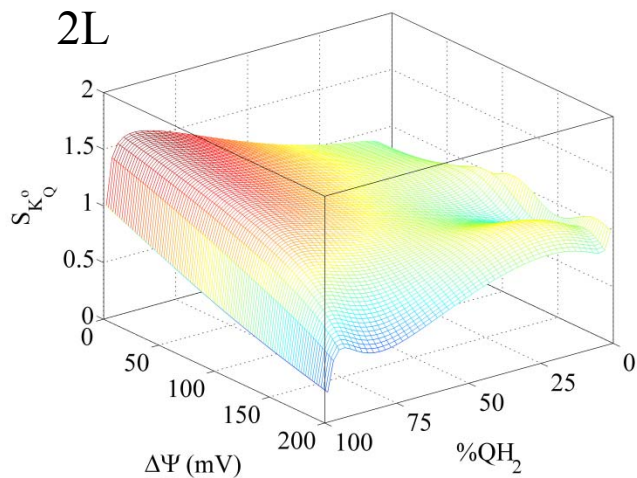
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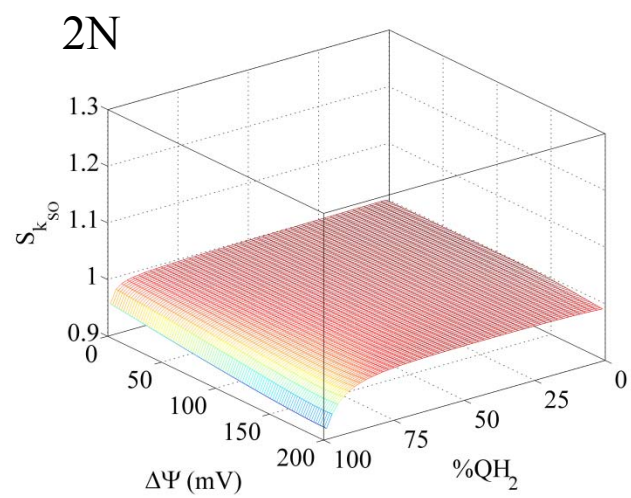
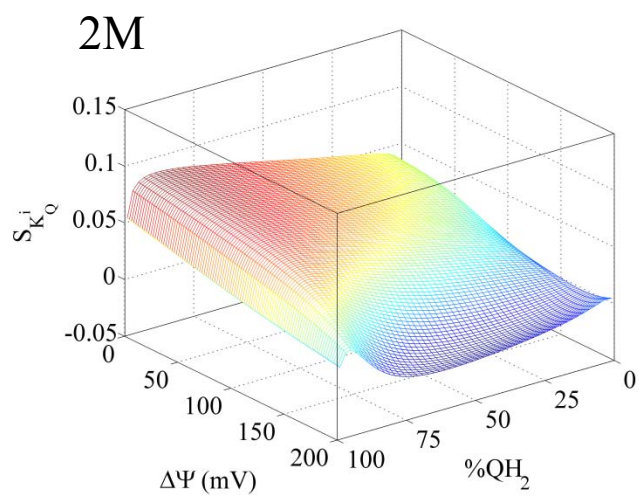
2K



2L



Normalized Sensitivity Coefficients for J_{ROS}



SUPPORTING REFERENCES

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